

The Four Giants as Balanced Pairs: Throats, Hearts, the Gas Ladder and the Interstellar Visitors

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Abstract

We show that each of the four giant planets is built of balanced pairs about a centre at four levels: its throats, its heart, its cloud decks and its place among the giants; the interstellar visitors and the planets of a pulsar meet the same structure. Every giant's Hill sphere opens through two throats, at L_1 and L_2 , whose linear dynamics is the same for every planet, $\lambda^4 - 2\lambda^2 - 27 = 0$ ($\lambda^2 - \omega^2 = 2$, $\lambda^2\omega^2 = 27$, vertical frequency 2). The pair is unequal by $\mathcal{A} = \frac{h}{3}(1 - \frac{h^2}{27} + O(h^3))$, $h = (\mu/3)^{1/3}$, and the full problem gives 0.99972 to 1.00000 of $h/3$ for the four giants and the planets of PSR B1257+12. Near each throat the energy surface has the topology $S^2 \times I$ of a spatial slice of the Einstein–Rosen bridge, in phase space; the flux through it is the action $2\pi\Delta E/\omega$ of its Lyapunov orbit, to 1.8×10^{-6} at Jupiter's L_1 , and every long temporary capture by Jupiter in the Ohtsuka catalogue entered and left through it. Jupiter and Saturn have dilute cores, and both carry tracking signals read as p-modes; the $n = 1$ polytrope gives Jupiter's acoustic spacing to 4.5% from mass and radius alone, and in polytropes matched to each planet's J_2/q_ω the 10^5 -bar level lies at 3569 and 8786 km, where the winds inferred from gravity end (3000 and 9000 km). An equilibrium cloud-condensation model anchored at the 1-bar temperatures gives 3, 3, 4, 4 cloud decks. The NH_4SH titration is an exact two-state pair, $x_{\text{NH}_3} = ke^{\varepsilon/2}$, $x_{\text{H}_2\text{S}} = ke^{-\varepsilon/2}$, whose survivor above the deck is $\text{sign}(N - S)$. With deep $N/S = 4.73, 3.16, 0.195$ and 0.170 this resolves why the ice giants show H_2S and not NH_3 above their clouds: the model's saturated H_2S at the cloud tops is 0.26–1.8 ppm on Uranus, which brackets the detected 0.4–0.8 ppm, and 0.49–2.25 ppm on Neptune, which overlaps the detected 1–3 ppm.

The triangle of Jupiter–Saturn conjunctions turns one third of a turn in exactly one period of the great inequality, 883 years. The four orbit planes precess in three live modes and one exact zero mode, the invariable plane. The triangular points are stable by the Gascheau–Routh criterion, with their spectrum on the imaginary axis in pairs of opposite Krein signature that leave it only by colliding, and Trojans are known at all four giants. The outermost moons of Jupiter, Saturn and Neptune reach 0.633, 0.625 and 0.654 of the Hill radius and those of Uranus 0.407; we predict retrograde moons 42–51 million km from Uranus. The three interstellar objects share neither a direction nor a kinematic origin (largest pairwise velocity difference 75.5 km/s). 3I/ATLAS touched Jupiter's Hill sphere, at 1.0011 Hill radii (1.0020 by JPL), and was not the source of the Wow! signal: in 1977 it lay 8.3° from the nearer beam centre, and a hydrogen-line transmitter on it would have been received 11.6 channels from the Wow!. Its four OH lines are two pairs about one centre, $1612.2309 + 1720.5299 = 1665.4018 + 1667.3590$ MHz. PSR B1257+12 holds a near-3:2 pair of planets whose angular momenta stand in the ratio 1.03 ± 0.07 , and for its mass the horizon radius is exactly one third of the innermost stable orbit. The results rest on exact expansions and linear algebra of the restricted problem, on a thermodynamic model built from stated data, and on two-body propagation of the visitors against an ephemeris that reproduces JPL's approach of 3I/ATLAS to Jupiter to 3×10^{-4} au.

1 Introduction

The four giant planets have been measured closely in the last decade. Juno’s radio tracking gives Jupiter’s gravity field and Cassini’s Grand Finale gave Saturn’s [31, 60]; interior models fitted to these fields place a dilute core in Jupiter and a stably stratified core in Saturn [151, 101, 92]; both tracking records carry signals read as normal modes [32, 93]; and hydrogen sulfide has been detected above the clouds of Uranus and Neptune [61, 62]. Three interstellar objects have crossed the planetary region since 2017 [95, 102, 103]. The theory this paper applies to them is classical, and its main steps form a sequence (Table 1).

year	authors	result
1843, 1874	Gascheau; Routh [42, 121]	the equilateral configuration is linearly stable iff $27 \sum_{i < j} m_i m_j < (\sum_i m_i)^2$
1968, 1969	Conley; McGehee [20, 94]	near a collinear point the planar energy surface is $S^2 \times I$
1969, 1973	Lewis; Weidenschilling, Lewis [79, 153]	equilibrium cloud-condensation model of the giant planets
1990	MacKay [88]	the flux over a saddle equals the action of its periodic orbit
1992, 2003	Wolszczan, Frail; Konacki, Wolszczan [163, 74]	the planets of PSR B1257+12; their masses and inclinations
1999	Cabral, Schmidt [15]	stability of a ring of N vortices with a centre
2005	Atreya, Wong [6]	coupled clouds and chemistry; the NH_4SH equilibrium constant
2018, 2019	Irwin et al. [61, 62]	H_2S above the clouds of Uranus and Neptune
2021	Molter et al. [105]	deep $\text{N/S} = 0.20$ in Uranus
2021	Mankovich, Fuller [92]	a diffuse, stably stratified core in Saturn from ring seismology

Table 1: Prior results this paper builds on.

Companion papers. The restricted problem treated here is the three-body problem of the physics companion [135] with one vanishing mass: its Lagrange triangle, which grows at exactly $\omega/\sqrt{2}$ for equal masses, is here the pair of triangular points stable by the Gascheau–Routh criterion [42, 121], and its figure-eight holds the triad $(-s, 0, s)$ that the throats hold as a pair about a centre. The chemistry companion [137] derives the two-state law of a metal-oxide gas sensor, $p = 1/(1 + e^\epsilon)$, and measures the sensor ladder of an urban air record; the NH_4SH deck of Section 5 obeys the same law, and Section 5.9 states exactly what is shared. The mathematics and quantum companions [134, 136] show that the zeros of ξ lie on the critical line exactly when each is its own mirror image, and that a positive merge of spins keeps every Lee–Yang zero on its circle while a negative coupling lets zeros leave in mirrored pairs; the spectrum at the triangular points (Theorem 5) carries the same Klein four-group as these zeros, with the imaginary axis as its fixed line, and its eigenvalues leave that line only where two pairs of opposite Krein signature meet. The survivor $\text{sign}(N - S) \in \{-1, 0, 1\}$ of the titration is a balanced-ternary digit [138]. The synthesis paper [139] collects these readings of a zero.

1.1 Main results

1. *Two throats about one centre* (Sections 3.1 and 3.2). The collinear points of Hill’s problem have the characteristic polynomial $\lambda^4 - 2\lambda^2 - 27 = 0$ for every planet, so $\lambda_H^2 - \omega_H^2 = 2$ and $\lambda_H^2 \omega_H^2 = 27$, with vertical frequency 2. The two throats are unequal by $\mathcal{A} = \frac{h}{3}(1 - \frac{h^2}{27} + O(h^3))$, $h = (\mu/3)^{1/3}$ (Theorem 1). In the full restricted problem $\mathcal{A}/(h/3)$ is 0.99972

- for Jupiter and within 10^{-5} of 1 for the planets of PSR B1257+12, and the largest relative departure of a linear rate from its Hill value is $0.96h$ to $1.01h$ (Table 3).
2. *The throat in phase space* (Section 3.3). Near each throat the planar energy surface is $S^2 \times I$, the topology of a spatial slice of the Einstein–Rosen bridge. The flux through the throat is the action $2\pi\Delta E/\omega$ of its Lyapunov orbit, which the computed orbits at Jupiter’s L_1 approach monotonically, to 1.000002 times it at amplitude $0.0018r_H$. The quantum transmission through the saddle is the two-state law $T = 1/(1 + e^{-2\pi\Delta E/\hbar\lambda})$, equal to $\frac{1}{2}$ at the saddle energy. Every long temporary capture of a comet by Jupiter in the Ohtsuka catalogue entered and left through the L_1 or L_2 region.
 3. *The hearts* (Section 4). The $n = 1$ polytrope has Love number $k_2 = 15/\pi^2 - 1$ and $J_2/q_\omega = 0.17327$ (Theorem 2), and all four giants lie below it, Saturn lowest at 0.10303. The same polytrope gives Jupiter’s acoustic spacing as $162.3\mu\text{Hz}$, 4.5% above the measured $155.3 \pm 2.2\mu\text{Hz}$, from mass and radius alone. Jupiter and Saturn hold their heavy elements in dilute cores and both carry tracking signals read as p-modes. In polytropes matched to each planet’s J_2/q_ω the 10^5 -bar level lies at 3569 and 8786 km, where the winds inferred from gravity end (3000 and 9000 km), and we predict Saturn’s acoustic spacing at 112–117 μHz . A ring of N equal vortices with a centre is spectrally stable exactly on the intervals of Cabral and Schmidt’s theorem, and three equal vortices with an equal fourth at the centre sit exactly on the edge.
 4. *The gas ladder* (Section 5). An equilibrium cloud-condensation model built from stated thermodynamic data and anchored at the 1-bar temperatures reproduces published cloud bases (NH_3 ice on Jupiter within 1.5–2.0% of Atreya and Wong) and gives 3, 3, 4, 4 cloud decks (Table 8). The NH_4SH titration is an exact two-state pair (Theorem 3): where the solid coexists with the gas, $x_{\text{NH}_3} = ke^{\varepsilon/2}$ and $x_{\text{H}_2\text{S}} = ke^{-\varepsilon/2}$ with $k^2 = K(T)/P^2$, the odd part $N - S$ is conserved, and the survivor above the deck is $\text{sign}(N - S)$. The deep ratios $N/S = 4.73, 3.16, 0.195$ and 0.170 of Jupiter, Saturn, Uranus and Neptune therefore leave NH_3 above the gas giants and H_2S above the ice giants. This resolves why the ice giants show H_2S and not NH_3 above their clouds: the model’s saturated H_2S at the cloud tops is 0.26–1.8 ppm on Uranus, which brackets the detected 0.4–0.8 ppm, and 0.49–2.25 ppm on Neptune, which overlaps the detected 1–3 ppm. The deck and a metal-oxide gas sensor obey one two-state law (Section 5.9).
 5. *The trigon* (Theorem 4). Each corner of the triangle of Jupiter–Saturn conjunctions turns at $\frac{1}{3}(5n_S - 2n_J)$, so the trigon advances one third of a turn in exactly one period of the great inequality, 883 years.
 6. *The zero mode* (Theorem 6). The Laplace–Lagrange inclination matrix of the four giants has the exact eigenvalue 0, whose invariant plane is the invariable plane, and three live modes, $-25.5183, -2.9276$ and -0.6698 arcsec/yr.
 7. *Gascheau–Routh and Krein* (Sections 6.4, 6.6 and 6.7). The triangular points are stable exactly for $\mu < \mu_G = 0.0385209$, and a stable triangle needs a dominant centre, $m_1 > 0.9614791M$. Below μ_G the spectrum at L_4 lies on the imaginary axis in two pairs of opposite Krein signature, which leave the axis only by colliding at $\omega = 1/\sqrt{2}$ (Theorem 5). All four giants satisfy the criterion, and Trojans are known at all four. Every orbit tangent to two neighbouring giants lies below both throat energies (Theorem 7). The outermost moons of Jupiter, Saturn and Neptune reach 0.633, 0.625 and 0.654 of the Hill radius and those of Uranus 0.407, and we predict undiscovered retrograde moons of Uranus 42–51 million km from the planet.
 8. *The visitors* (Section 7). The three interstellar objects share neither a direction nor a kinematic origin: no statistic of their incoming directions departs from isotropic or kinematic populations (smallest $p = 0.27$), and their largest pairwise velocity difference, 75.5 km/s, lies at the median, 75.4 km/s, of random triples from the kinematic population. 3I/ATLAS touched Jupiter’s Hill sphere on 16 March 2026, at 1.0011 Hill radii

(1.0020 by JPL). 3I/ATLAS was not the source of the Wow! signal: in 1977 it lay 8.3° from the nearer beam centre, and a hydrogen-line transmitter on it would have been received 11.6 channels from the Wow!. Its four OH lines are two pairs about one centre, $1612.2309 + 1720.5299 = 1665.4018 + 1667.3590$ MHz, and its OH lines changed from absorption to emission between heliocentric radial velocities $+13.37$ and $+20.93$ km/s, where the inversion crosses zero.

9. *PSR B1257+12* (Section 8). The pulsar’s planets B and C carry the giants’ two throats, unequal by $h/3$ to 10^{-5} , lie 1.60% inside the 3:2 resonance and carry angular momenta in the ratio $L_C/L_B = 1.03 \pm 0.07$. For a mass M the horizon, the photon sphere and the innermost stable circular orbit stand at 2, 3 and 6 in units of GM/c^2 , so the horizon radius is exactly one third of the innermost stable orbit.

1.2 Outline of the method

The throat results are exact expansions of the collinear equilibrium conditions and linear algebra of the restricted problem; the full-problem values come from root finding, and the flux from Lyapunov orbits computed by differential correction. The hearts use the exact Love number of the $n = 1$ polytrope, first-order Clairaut theory on Lane–Emden profiles and the analytic Jacobian of the $N + 1$ vortex problem. The gas ladder integrates the dry adiabat of normal hydrogen, with heat capacities from the rotational levels of H_2 , by the fourth-order Runge–Kutta method in $\ln P$ from the NSSDC 1-bar temperatures, and locates each deck base as the balance point of its saturation ratio by root bracketing; the titration theorem follows from mass action and one-for-one consumption. The results on the set follow from the mean motions, the Laplace–Lagrange matrix and the linearisation at L_4 , with Krein signatures computed from the quadratic Hamiltonian. The visitors are propagated on their two-body hyperbolae against planetary positions from the orbital elements and periodic terms of Schlyter’s method [128], rotated to the J2000 ecliptic; this ephemeris reproduces JPL’s closest approach of 3I/ATLAS to Jupiter, 0.358330 au at 12:22 UT on 16 March 2026, as 0.358011 au at 11:33 UT (Section 7.2). Their directions and velocities are compared with isotropic and kinematic populations by Monte Carlo sampling (10^6 draws).

1.3 Organisation and notation

Section 2 collects the planetary data. Section 3 treats the throats, Section 4 the hearts, Section 5 the gas ladder, Section 6 the giants as a set, Section 7 the visitors and Section 8 the planets of PSR B1257+12; Section 9 sets the structure out level by level. Every computed number is obtained from the stated closed forms, by root finding, by numerical linear algebra, by numerical integration of the stated equations, or by Monte Carlo sampling of the stated populations, by the method given where the number appears; arithmetic is IEEE double precision. Frequencies in hertz are written ν ; dimensionless frequencies in a rotating frame, in units of the mean motion, are written ω . N and S are the deep mole fractions of NH_3 and H_2S , and r_H is the Hill radius.

2 The four giants

Table 2 collects the data used throughout. The rotation parameter is $q_\omega = \omega_{\text{rot}}^2 R^3 / GM$ at the reference radius R of each gravity field, and J_2/q_ω measures how strongly mass gathers towards the centre (Section 4.1). Hubble’s auroral record gives Uranus’s day as 17.247864 h, 28 s longer than Voyager’s value [76]; we use it. Uranus’s obliquity is 97.77° and Neptune’s 28.32° ; their magnetic dipoles are tilted by 58.6° and 46.9° and offset by 0.352 and 0.485 planetary radii

[162, 160]. By the counts of April and June 2026, Jupiter has 115 known moons and Saturn 293 with confirmed orbits; Uranus has 29, including S/2025 U 1, and Neptune 16 [155].

	GM ($10^6 \text{ km}^3 \text{s}^{-2}$)	a (10^6 km)	R (km)	J_2 (10^{-6})	rotation	q_ω	J_2/q_ω	I/MR^2
Jupiter	126.687	778.479	71492	14696.5735	9.9250 h	0.08919	0.16477	0.2582
Saturn	37.931	1432.041	60330	16290.573	$10^{\text{h}}33^{\text{m}}38^{\text{s}}$	0.15812	0.10303	0.2189
Uranus	5.7940	2867.043	25559	3343.43	17.247864 h	0.02951	0.11331	0.2259
Neptune	6.8351	4514.953	24764	3411	16.11 h	0.02608	0.13080	0.2374

Table 2: The four giants. GM , a and the rotation of Jupiter and Neptune from the NSSDC fact sheets [159, 161, 162, 160]; J_2 of Jupiter from Juno [31] and of Saturn from Cassini [60], at the reference radii shown; J_2 of Uranus and Neptune from NSSDC at the equatorial radius; Saturn’s rotation from ring seismology [91]; Uranus’s from [76]. The $n = 1$ polytrope has $J_2/q_\omega = 0.17327$ (Theorem 2); smaller values mean more mass towards the centre. Last column: the first-order Radau–Darwin moment of inertia (Section 4.1).

3 Two throats about one centre

3.1 Hill’s problem and the universal quartic

In the frame turning with a planet of mass fraction $\mu = m/(M + m)$ about a star of mass M , with the separation, the mean motion n and $G(M + m)$ set to one, a test body keeps the Jacobi constant

$$C = 2U - v^2, \quad U = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2},$$

where r_1, r_2 are its distances from the star and the planet [107]. Near the planet, lengths scale with $\mu^{1/3}$; in these units (Hill’s units) and in the limit $\mu \rightarrow 0$ the equations of motion become Hill’s equations,

$$\ddot{x} - 2\dot{y} = 3x - \frac{x}{r^3}, \quad \ddot{y} + 2\dot{x} = -\frac{y}{r^3}, \quad \ddot{z} = -z - \frac{z}{r^3},$$

which are the same for every planet. They derive from the effective potential $U_H = \frac{3}{2}x^2 - \frac{1}{2}z^2 + 1/r$. On the x -axis, $\partial_x U_H = 3x - x/r^3 = 0$ gives the two collinear points $x = \pm 3^{-1/3} = \pm 0.693361$, where $r^3 = \frac{1}{3}$. There $\partial_{xx}^2(1/r) = 2/r^3$ and $\partial_{yy}^2(1/r) = \partial_{zz}^2(1/r) = -1/r^3$, so

$$U_{xx} = 3 + \frac{2}{r^3} = 9, \quad U_{yy} = -\frac{1}{r^3} = -3, \quad U_{zz} = -1 - \frac{1}{r^3} = -4.$$

The linearised planar equations $\ddot{x} - 2\dot{y} = U_{xx}x$, $\ddot{y} + 2\dot{x} = U_{yy}y$ have solutions $e^{\lambda t}$ when $\det \begin{pmatrix} \lambda^2 - U_{xx} & -2\lambda \\ 2\lambda & \lambda^2 - U_{yy} \end{pmatrix} = 0$, that is

$$\lambda^4 + (4 - U_{xx} - U_{yy})\lambda^2 + U_{xx}U_{yy} = \lambda^4 - 2\lambda^2 - 27 = 0. \quad (1)$$

Hence $\lambda^2 = 1 \pm 2\sqrt{7}$: a real pair $\pm\lambda_H$, $\lambda_H = \sqrt{1 + 2\sqrt{7}} = 2.508287$, and an imaginary pair $\pm i\omega_H$, $\omega_H = \sqrt{2\sqrt{7} - 1} = 2.071594$; the vertical motion adds $\pm 2i$. The two roots in λ^2 have sum 2 and product -27 , so

$$\lambda_H^2 - \omega_H^2 = 2, \quad \lambda_H^2 \omega_H^2 = 27 = 3^3.$$

The numbers 2 and 3^3 fix every planet’s throat. Departures from the saddle grow by a factor e in $1/\lambda_H$ of the time unit $1/n$, that is in $P_{\text{orb}}/(2\pi\lambda_H) = P_{\text{orb}}/15.7600$ of the planet’s orbital period. The eigenvalues of the Jacobian of Hill’s equations agree with these roots to 4.4×10^{-16} .

3.2 The full problem: every giant carries the same pair

In the full restricted problem we located L_1 and L_2 by root finding and computed the eigenvalues of the linearisation for the four giants and for the three planets of PSR B1257+12 (Section 8). Table 3 lists them. For every body the largest departure of a rate from its Hill value lies between $0.96h$ and $1.01h$, with $h = (\mu/3)^{1/3}$. In the full problem the e -folding time at Jupiter's L_1 is 257.26 days.

	μ	r_H (Mkm)	L_1, L_2 (Mkm)	$\mathcal{A}/(h/3)$	λ at L_1, L_2	$C(L_1)$	libr. (yr)
Jupiter	9.537×10^{-4}	53.130	51.906, 54.322	0.99972	2.681, 2.352	3.038756	147.4
Saturn	2.857×10^{-4}	65.399	64.392, 66.383	0.99989	2.622, 2.402	3.017821	670.2
Uranus	4.366×10^{-5}	69.997	69.423, 70.562	0.99997	2.568, 2.451	3.005219	4893
Neptune	5.150×10^{-5}	116.471	115.462, 117.465	0.99997	2.572, 2.447	3.005818	8837
PSR B	9.225×10^{-6}	0.7821	0.7783, 0.7858	0.99999	2.544, 2.474	3.001872	23.09
PSR C	8.367×10^{-6}	0.9814	0.9767, 0.9859	0.99999	2.543, 2.475	3.001755	35.78

Table 3: The throats in the full restricted problem. $\mu = m/(M + m)$ and $r_H = a(\mu/3)^{1/3}$, from the NSSDC values of Table 2; the pulsar planets from the timing solution of [74] with a $1.4 M_\odot$ pulsar (Section 8). Section 6.7 uses $r_H = a(m/3M_\odot)^{1/3}$, 0.03% larger at Jupiter. $\mathcal{A} = (\gamma_2 - \gamma_1)/(\gamma_2 + \gamma_1)$, with $\gamma_{1,2}$ the distances of $L_{1,2}$ from the planet; λ in units of the mean motion (Hill value 2.5083); $C(L_1)$ the Jacobi constant at L_1 ; last column, the Trojan libration period at L_4 (Section 6.4).

Proposition 1 (the pair is unequal by a third). *Let γ_1, γ_2 be the distances of L_1 and L_2 from the planet and $h = (\mu/3)^{1/3}$. Then*

$$\gamma_1 = h - \frac{h^2}{3} - \frac{h^3}{9} + \frac{58}{81}h^4 + O(h^5), \quad \gamma_2 = h + \frac{h^2}{3} - \frac{h^3}{9} + \frac{50}{81}h^4 + O(h^5),$$

and

$$\mathcal{A} = \frac{\gamma_2 - \gamma_1}{\gamma_2 + \gamma_1} = \frac{h}{3} \left(1 - \frac{h^2}{27} + O(h^3) \right).$$

Proof. Place the star at $x = -\mu$ and the planet at $x = 1 - \mu$. At L_1 , $x = 1 - \mu - \gamma$, the centrifugal term, the star's pull and the planet's pull balance: $1 - \mu - \gamma - (1 - \mu)(1 - \gamma)^{-2} + \mu\gamma^{-2} = 0$. Expanding $(1 - \gamma)^{-2} = 1 + 2\gamma + 3\gamma^2 + 4\gamma^3 + 5\gamma^4 + \dots$, the constant terms cancel and $\mu = \gamma^3 + (1 - \mu)(2\gamma^3 + 3\gamma^4 + 4\gamma^5 + 5\gamma^6 + \dots)$. Since $\mu = 3\gamma^3(1 + O(\gamma))$, the term $-2\mu\gamma^3$ equals $-6\gamma^6 + O(\gamma^7)$, and

$$\frac{\mu}{3} = \gamma^3 \left(1 + \gamma + \frac{4}{3}\gamma^2 - \frac{1}{3}\gamma^3 + O(\gamma^4) \right).$$

At L_2 , $x = 1 - \mu + \gamma$, the balance is $1 - \mu + \gamma - (1 - \mu)(1 + \gamma)^{-2} - \mu\gamma^{-2} = 0$, and the same steps with $(1 + \gamma)^{-2} = 1 - 2\gamma + 3\gamma^2 - 4\gamma^3 + 5\gamma^4 - \dots$ give $\frac{\mu}{3} = \gamma^3 \left(1 - \gamma + \frac{4}{3}\gamma^2 - \frac{11}{3}\gamma^3 + O(\gamma^4) \right)$. With $\mu/3 = h^3$ and $(1 + u)^{1/3} = 1 + \frac{u}{3} - \frac{u^2}{9} + \frac{5u^3}{81} + O(u^4)$,

$$h = \gamma \left(1 + \frac{\gamma}{3} + \frac{\gamma^2}{3} - \frac{28}{81}\gamma^3 \right) \text{ at } L_1, \quad h = \gamma \left(1 - \frac{\gamma}{3} + \frac{\gamma^2}{3} - \frac{80}{81}\gamma^3 \right) \text{ at } L_2,$$

up to $O(\gamma^5)$. Substituting $\gamma = h + c_2h^2 + c_3h^3 + c_4h^4$ and equating powers of h gives $c_2 = \mp \frac{1}{3}$, $c_3 = -\frac{1}{9}$ and $c_4 = \frac{58}{81}, \frac{50}{81}$ at L_1, L_2 , which are the stated expansions. Hence $\gamma_2 - \gamma_1 = \frac{2}{3}h^2 - \frac{8}{81}h^4 + O(h^5)$ and $\gamma_2 + \gamma_1 = 2h - \frac{2}{9}h^3 + O(h^4)$, so

$$\mathcal{A} = \frac{h}{3} \cdot \frac{1 - \frac{4}{27}h^2 + O(h^3)}{1 - \frac{1}{9}h^2 + O(h^3)} = \frac{h}{3} \left(1 - \frac{4}{27}h^2 + \frac{1}{9}h^2 + O(h^3) \right) = \frac{h}{3} \left(1 - \frac{h^2}{27} + O(h^3) \right).$$

□

The throats are a pair about the planet, L_1 on the star side and L_2 outside, unequal by exactly one third of h at first order. The computed $\mathcal{A}/(h/3)$ is 0.99972 for Jupiter and within

10^{-5} of 1 for the pulsar planets (Table 3); $|\mathcal{A}/(h/3) - 1|$ never exceeds $0.0041h$. Solving the equilibrium conditions in 50-digit arithmetic for $\mu = 10^{-9}$ and 10^{-6} , $\gamma_{1,2}/h$ agree with the expansions of Theorem 1 to $0.2h^4$, and $(\mathcal{A}/(h/3) - 1)/h^2 = -0.037268$ and -0.039351 , which are $-\frac{1}{27} - \frac{h}{3}$ to 3×10^{-8} and 3×10^{-6} : the next coefficient is $-\frac{1}{3}$. For Jupiter, $1 - \frac{h^2}{27} - \frac{h^3}{3} = 0.99972$, the value in Table 3.

3.3 The throat in phase space

For Jacobi constants between $C(L_2)$ and $C(L_1)$ only the L_1 throat is open; between $C(L_3)$ and $C(L_2)$ both are (for Jupiter, $3.000954 < C < 3.037484$). The zero-velocity surface $2U = C$ is closed everywhere else, so a body can enter or leave the planet's realm only through the throats (Fig. 1a). Conley and McGehee showed that near each throat the planar energy surface is a sphere times an interval, $S^2 \times I$ [20, 94]. In the spatial problem it is $S^4 \times I$; it contains an invariant 3-sphere, whose stable and unstable manifolds are tubes $S^3 \times \mathbb{R}$ that separate transit from non-transit orbits [75, 46]. A spatial slice through the Schwarzschild wormhole of Einstein and Rosen [34] is $S^2 \times \mathbb{R}$ [104]. Near each throat, therefore, the energy surface has the topology $S^2 \times I$ of an Einstein–Rosen slice, in phase space, and the tubes play the part of the bridge: they carry every orbit that passes from one realm to the other. A wormhole in spacetime would also lens the light of stars behind it and pull on nearby bodies through its mass; the appearance of such a wormhole near Saturn was computed from general relativity for the film *Interstellar* [67]. Cassini's radio ranging to Saturn is precise enough to constrain where an unseen ninth planet could lie far beyond it [38], and the one signal in its tracking that the gravity field did not absorb is read as Saturn's own oscillations (Section 4.3).

The flux through the throat, and its quantum form. In planar normal coordinates near the throat, $H = \lambda q_1 p_1 + \frac{\omega}{2}(q_2^2 + p_2^2)$. At energy ΔE above the saddle the periodic (Lyapunov) orbit is the circle $\frac{\omega}{2}(q_2^2 + p_2^2) = \Delta E$, $q_1 = p_1 = 0$, of radius $\sqrt{2\Delta E/\omega}$, and its action is the enclosed area,

$$\oint p dq = \pi \cdot \frac{2\Delta E}{\omega} = \frac{2\pi\Delta E}{\omega}.$$

MacKay proved that the flux of orbits across the dividing surface through a saddle equals the action of this periodic orbit [88]. For Lyapunov orbits computed by differential correction at Jupiter's L_1 , the ratio of the action to $2\pi\Delta E/\omega$ falls monotonically from 1.001274 (amplitude $0.044 r_H$) to 1.000002 ($0.0018 r_H$), the orbits closing to better than 1.1×10^{-12} . In chemistry this orbit is a reaction's transition state: Porter and Cvitanović call the orbits about L_1 and L_2 “the transition states ... controlling transport through the bottleneck” [119], and Jaffé and co-workers computed asteroid escape rates from them with the statistical rate theory of molecules [66]. Quantum mechanically, a mode crossing such a saddle with energy ΔE above it is transmitted with probability

$$T(\Delta E) = \frac{1}{1 + e^{-2\pi\Delta E/\hbar\lambda}}$$

(the parabolic barrier of [77, §50]), and the transverse channels open at $\Delta E = \hbar\omega(n + \frac{1}{2})$, so semiclassically the flux divided by $2\pi\hbar$ counts the open channels. T is the two-state law $p = 1/(1 + e^\varepsilon)$ of the chemistry companion [137] with log-odds $\varepsilon = -2\pi\Delta E/\hbar\lambda$: transmission and reflection are the two members of a pair, and they balance, $T = \frac{1}{2}$, exactly at the saddle energy.

Traffic through the throats. The throats themselves are not seen; their traffic is. Ohtsuka and co-workers tabulate the long temporary captures of comets by Jupiter [113]. Comet

147P/Kushida–Muramatsu was held from 1949 to 1961, entering near L_2 and leaving near L_1 ; 82P and P/1996 R2 went $L_2 \rightarrow L_1$, 111P $L_1 \rightarrow L_2$ over 18.45 years, and Shoemaker–Levy 9 came through L_1 and circled for more than 50 years before striking Jupiter. Every entry and exit listed passes through the L_1 or L_2 region. Comets Oterma and Gehrels 3 switch between orbits near the exterior 2:3 and interior 3:2 resonances by way of the throats [75]. The “arches of chaos”, invariant manifolds mostly of Jupiter’s collinear points, carry small bodies to Neptune’s distance in under a decade [145], and they are organised by the manifolds of the unstable periodic orbits of first-order resonances and their heteroclinic connections, alongside the classical throat channels [48].

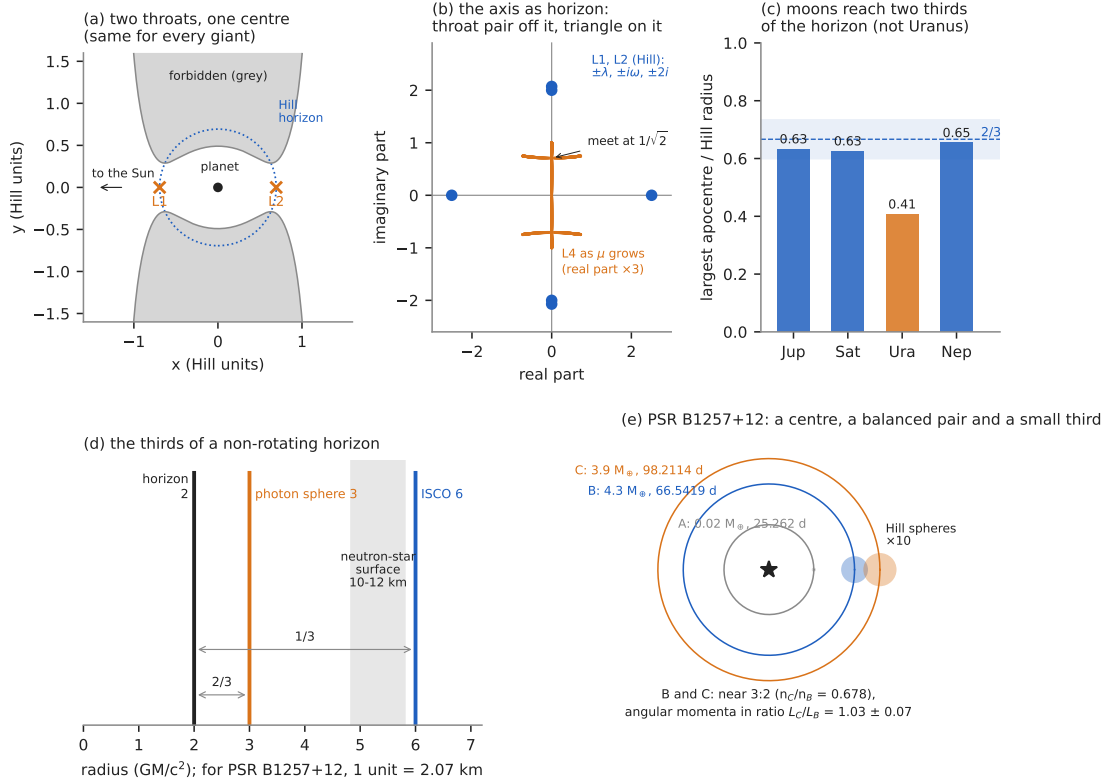


Figure 1: (a) The zero-velocity curve in Hill’s units at $C = 3^{4/3} - 0.25$, below the throat value: the planet’s realm opens towards the Sun at L_1 and outwards at L_2 ; dotted, the Hill radius. (b) Blue: the throat spectrum, a real pair $\pm\lambda$ off the imaginary axis and two pairs on it. Orange: the L_4 spectrum as μ grows from 10^{-5} to 0.06; the two pairs meet at $1/\sqrt{2}$ and leave the axis as a quadruple (real parts drawn three times larger). (c) The largest apocentre of each giant’s known moons against its Hill radius; the band marks 0.600–0.733, within 10% of $\frac{2}{3}$. (d) Schwarzschild radii in units of GM/c^2 ; grey, neutron-star radii of 10–12 km for $1.4 M_\odot$. (e) The planets of PSR B1257+12, with Hill spheres enlarged tenfold.

4 The hearts

Juno’s radio tracking gives Jupiter’s gravity field and its infrared mapper watches the poles [31, 32, 1]. Cassini’s Grand Finale in 2017 flew between Saturn and its rings and gave Saturn’s field [60]; earlier, its stellar occultations recorded 17 density waves and 4 bending waves in the C ring driven by Saturn’s own oscillations [91].

4.1 The rotational response

The response of J_2 to the spin measures how strongly mass gathers to the centre. For a uniform sphere $J_2/q_\omega = \frac{1}{2}$; for the $n = 1$ polytrope it follows exactly from the Love number.

Proposition 2 (the $n = 1$ polytrope). *For the polytrope $P = K\rho^2$ of mass M and radius R , $\rho = \rho_c \sin u/u$ with $u = \pi r/R$, $\rho_c = \pi M/4R^3$ and $K = 2GR^2/\pi$. Its fluid Love number is $k_2 = 15/\pi^2 - 1 = 0.519818$, and to first order in the rotation $J_2 = \frac{1}{3}k_2q_\omega$, so that $J_2/q_\omega = (15/\pi^2 - 1)/3 = 0.17327$.*

Proof. Let W be an applied potential with $\nabla^2 W = 0$ inside the body and Φ the self-potential. Hydrostatic equilibrium $\nabla P = -\rho \nabla(\Phi + W)$ with $P = K\rho^2$ gives $2K\rho \nabla \rho = -\nabla(\Phi + W)$, so $2K\rho + \Phi + W$ is constant, and Poisson's equation $\nabla^2 \Phi = 4\pi G\rho$ becomes $\nabla^2 \Phi + k^2 \Phi = k^2(c - W)$ with $k^2 = 2\pi G/K$. For $W = 0$ the regular solution gives $\rho \propto \sin(kr)/(kr)$, which first vanishes at $kR = \pi$; hence $K = 2GR^2/\pi$, and $M = 4\pi\rho_c(R/\pi)^3 \pi$ gives $\rho_c = \pi M/4R^3$. Now let $W = wr^2 P_2(\cos \theta)$ and write Φ_1 for the change of the self-potential. Inside, $\nabla^2 \Phi_1 + k^2 \Phi_1 = -k^2 W$, whose regular solution is $\Phi_1 = [-wr^2 + A j_2(kr)]P_2$, with j_2 the spherical Bessel function; outside, $\Phi_1 = Br^{-3}P_2$. The unperturbed density vanishes at $r = R$, so the displaced surface carries no mass at first order and Φ_1 and $\partial_r \Phi_1$ are continuous there. With $j_2(\pi) = 3/\pi^2$ and $j_2'(\pi) = j_1(\pi) - \frac{3}{\pi}j_2(\pi) = \frac{1}{\pi} - \frac{9}{\pi^3}$, matching gives

$$-wR^2 + A\frac{3}{\pi^2} = BR^{-3}, \quad -2wR + A\frac{\pi}{R}\left(\frac{1}{\pi} - \frac{9}{\pi^3}\right) = -3BR^{-4}.$$

Adding $3/R$ times the first equation to the second eliminates B and leaves $-5wR + A/R = 0$, so $A = 5wR^2$ and $BR^{-3} = wR^2(15/\pi^2 - 1)$: the response at the surface is $k_2 = 15/\pi^2 - 1$ times the applied potential. For uniform rotation the centrifugal potential $-\frac{1}{2}\omega_{\text{rot}}^2 r^2 \sin^2 \theta$ has quadrupole part $w = \omega_{\text{rot}}^2/3$. Outside, the response $k_2 \frac{\omega_{\text{rot}}^2}{3} R^5 r^{-3} P_2$ is the J_2 term $GM J_2 R^2 r^{-3} P_2$ of the exterior potential $-\frac{GM}{r}[1 - J_2(\frac{R}{r})^2 P_2]$, so $J_2 = k_2 \omega_{\text{rot}}^2 R^3 / 3GM = \frac{1}{3}k_2 q_\omega$. $\square$

All four giants lie below 0.17327 (Table 2): Jupiter at 0.16477, Saturn at 0.10303, Uranus at 0.11331 and Neptune at 0.13080. Saturn is the most centrally condensed. The polytropic index for which first-order Clairaut theory on the Lane–Emden profile reproduces the measured J_2/q_ω is 1.04 for Jupiter and 1.44 for Saturn. The first-order Radau–Darwin relation, $I/MR^2 = \frac{2}{3}[1 - \frac{2}{5}\sqrt{\eta - 1}]$ with $\eta = 5q_\omega/(3J_2 + q_\omega)$ [100], gives the moments of inertia in Table 2, against $\frac{3}{5}$ for a uniform sphere; it assumes a nearly uniform density and is least reliable for strongly layered planets.

4.2 The heart of Jupiter

Juno measured $J_2 = 14696.5735$, $J_4 = -586.6085$ and $J_6 = 34.2007$ ($\times 10^{-6}$, reference radius 71492 km) [31]. Interior models fitted to Juno's field place a dilute core between 0.3 and 0.5 of the radius [151]. In the five-layer reference model of Militzer and Hubbard the dilute core extends to 0.42 and a stably stratified transition layer to 0.62 of the radius, spanning the cumulative mass fractions 0.20 to 0.54; any compact core weighs at most 2.5 (rock) to 3.0 (rock–ice) Earth masses of the 23.5–25.5 Earth masses of heavy elements [101]. The zonal winds reach about 3000 km down, 4% of the radius [72, 73].

The heartbeat. Juno's Doppler record carries residuals of up to 0.1 mm/s over 10–15 minutes in its 22 gravity passes, which resist a low-order static interpretation; they are fitted by pressure modes (p-modes) at 900–1200 μHz with surface speeds of 10–50 cm/s and displacements of 15–80 m [32]. From the ground, Jupiter's oscillations peak at $1213 \pm 50 \mu\text{Hz}$ with a spacing of

$155.3 \pm 2.2 \mu\text{Hz}$ [43]. The spacing is set by the sound-crossing time, $\Delta\nu = [2 \int_0^R dr/c]^{-1}$. In the $n = 1$ polytrope with $\Gamma_1 = 2$, Theorem 2 gives

$$c^2 = \Gamma_1 \frac{P}{\rho} = \Gamma_1 K \rho = \frac{\Gamma_1 GM}{2R} \frac{\sin u}{u}, \quad \int_0^R \frac{dr}{c} = \frac{R}{\pi} \left(\frac{2R}{\Gamma_1 GM} \right)^{1/2} \int_0^\pi \left(\frac{u}{\sin u} \right)^{1/2} du,$$

and the last integral is 5.893400 by quadrature. With $\Gamma_1 = 2$, $\Delta\nu = \frac{\pi^2}{5.893400} \nu_{\text{dyn}} = 1.6747 \nu_{\text{dyn}}$, where $\nu_{\text{dyn}} = (GM/R^3)^{1/2}/2\pi$ is the planet's dynamical frequency. For Jupiter ($R = 69911$ km, the volumetric mean radius) the crossing takes 3081 s and $\Delta\nu = 162.3 \mu\text{Hz}$, 4.5% above the measurement, from mass and radius alone.

4.3 The heart of Saturn and the signal in the tracking

Cassini measured $J_2 = 16290.573$, $J_4 = -935.314$ and $J_6 = 86.340$ ($\times 10^{-6}$, reference radius 60330 km) [60]. The same tracking held a signal that the gravity solution could not absorb: range-rate residuals of up to 0.2 mm/s over 20–60 minutes, accelerations of order 10^{-7} m/s², handled in the published field by random accelerations in 10-minute steps near closest approach [60]. A forward model of Saturn's normal modes reproduces it, favouring p-modes at 500–700 μHz with amplitudes of several kilometres; no single fundamental mode had more than a 2% chance of producing it, and the corresponding signal at Jupiter is about 20 times weaker [93]. This is the gravitational anomaly in the data around Saturn, and its reading is Saturn's own ringing, recorded by the spacecraft that flew through it.

In the rings. The ring-wave pattern speeds, known to better than 0.1 deg/day, fixed Saturn's day at $10^{\text{h}}33^{\text{m}}38^{\text{s}}$, with the 5% and 95% quantiles $1^{\text{m}}19^{\text{s}}$ below and $1^{\text{m}}52^{\text{s}}$ above [91]. The gravity-mode waves fixed its heart: a stably stratified core out to 0.59 ± 0.01 of the radius (linear composition profile) or 0.70 ± 0.02 (smooth profile), half of the heavy elements within 0.32 ± 0.01 , and 19.1 ± 1.0 Earth masses of heavy elements, 17.4 ± 1.2 of them in the stable region [92].

4.4 What links the two hearts

One pressure. Jupiter's winds are inferred to end near 3000 km and Saturn's near 9000 km, and both depths lie near 10^5 bar [73], where hydrogen's conductivity reaches about 1 S/m and the magnetic field is expected to brake the flow [73, 41]. In the $n = 1$ polytrope of Theorem 2 ($\rho_c = 4363$ and 2260 kg/m³) the pressure $P = K\rho^2$ is 7.899×10^4 bar at 3000 km in Jupiter and 2.273×10^5 bar at 9000 km in Saturn, and the 10^5 bar level lies at 3360 and 6180 km. In polytropes whose first-order J_2/q_ω matches each planet ($n = 1.04$ and 1.44) the pressures are 6.908×10^4 and 1.070×10^5 bar, and the 10^5 bar level lies at 3569 and 8786 km (Fig. 2b); the mid-point of Saturn's inferred wind decay is at 8743 km [41]. Saturn's greater central condensation therefore brings the two depths to one pressure: the ratio of the pressures falls from 2.88 ($n = 1$) to 1.55. The depth ratio of the 10^5 bar level, 2.46, follows mostly from the surface gravities, 24.79 and 10.44 m/s² (ratio 2.37), since layers of one material reach a given pressure at depths in the inverse ratio of g . One physical level, found in two planets at depths near the inverse ratio of their gravities, is the equivalence that links their winds.

One heartbeat. Both planets carry tracking signals best read as p-modes [32, 93]. Each planet's natural clock is ν_{dyn} : 96.91 μHz for Jupiter and 69.75 μHz for Saturn, ratio 1.3893. The ratios of the published p-mode bands (Juno 900–1200, Cassini 500–700 μHz) run from 1.286 to 2.400, so the bands are consistent with each planet ringing on its own clock. Scaled

from Jupiter’s measured spacing by ν_{dyn} , Saturn’s acoustic spacing is $111.8\mu\text{Hz}$; the $n = 1$ polytrope gives $116.8\mu\text{Hz}$. We predict Saturn’s spacing at $112\text{--}117\mu\text{Hz}$; a long, continuous record of Saturn’s disk or gravity would measure it.

One shape of heart. Both planets hold their heavy elements in a dilute, stratified core (Fig. 2a). In Saturn half of them lie within 0.32 ± 0.01 of the radius, 1.33σ from one third, and the stable region ends at 0.59 ± 0.01 or 0.70 ± 0.02 , 7.67σ below and 1.67σ above two thirds. In Jupiter the dilute core and the transition layer reach 0.42 and 0.62 .

4.5 The poles: a centre and its ring

Jupiter’s north pole has eight cyclones around one, its south pole five around one [1]. On 4 November 2019 a sixth joined the southern ring [108]; it was gone two months later [106]. The classical model is a ring of N point vortices of strength 1 at radius 1 around a central vortex of strength κ , rotating rigidly at $\Omega_N = ((N-1)/2 + \kappa)/2\pi$ [143]. We computed its linear stability, with the central vortex free to move, from the analytic Jacobian in the co-rotating frame, which agrees with finite differences to 9.55×10^{-10} (Fig. 2c). Thomson’s heptagon ($N = 7$, no centre) is neutral (largest growth rate 9.9×10^{-9}), and eight without a centre grow at 0.450 . For $N = 4, \dots, 12$ the computed intervals of spectral stability coincide, to 4×10^{-11} at both edges, with those of Cabral and Schmidt’s theorem [15, 9],

$$\frac{N^2 - 8N + 8}{16} < \kappa < \frac{(N-1)^2}{4} \quad (N \text{ even}), \quad \frac{(N-1)(N-7)}{16} < \kappa < \frac{(N-1)^2}{4} \quad (N \text{ odd}).$$

For $N = 3$ the upper edge is $\kappa = 1$ and no lower edge lies above -20 . At the upper edge the centre holds the share $\kappa/(\kappa + N) = (N-1)^2/((N-1)^2 + 4N) = ((N-1)/(N+1))^2$ of the total strength.

- *Three with an equal fourth sit on the edge.* For $N = 3$, $\kappa_{\text{max}} = 1$: an equal fourth vortex at the centre of three leaves the triad exactly marginal. The growth rate is zero below $\kappa = 1$, 5.0×10^{-3} at 1.001 and 1.6×10^{-2} at 1.01 , rising as $\sqrt{\kappa - 1}$.
- *Large rings need a centre.* For $N \geq 8$ the lower edge is positive. Jupiter’s northern $8 + 1$ with an equal centre has central share $\frac{1}{9}$, inside its interval $[0.059, 0.605]$ (largest growth rate 1.0×10^{-8}).
- *The south.* With equal centres, $5 + 1$ and $6 + 1$ are both inside their intervals (growth rates 6.6×10^{-9} and 1.9×10^{-16}), so the departure of the sixth cyclone is set by what the planar model leaves out: on the planet each cyclone is wrapped in a shielding ring of opposite vorticity [82], and the ring of cyclones sits on a curved pole at 83° , 8700 km from the centre [18].

Saturn’s poles and hemispheres. In the north, the hexagon’s vertices lie at 74.7° planetocentric latitude and turn in $10^{\text{h}}39^{\text{m}}23.01^{\text{s}}$; its 120 m/s jet survived the polar night [122]. In the south, a ten-sided wave formed at 63°S : faint in October 2023, sharper in August 2024, complete with ten vertices in August–September 2025 [123], in the months after the Sun crossed the ring plane on 6 May 2025 [4]; its winds are reported as about 400 km/h [147]. Saturn’s northern and southern radio emissions pulse with their own periods, about 10.6 and 10.8 hours before the 2009 equinox; the two converged, crossed shortly after the equinox of 11 August 2009, and merged near 10.7 hours in spring 2010 [39]. The opposite hemispheres, one on each side of the ring plane, joined in one period in the months after the Sun crossed that plane. Saturn’s magnetic dipole is tilted by less than 0.007° from its spin axis [16].

Jupiter’s poles in X-rays. Chandra’s High Resolution Camera found the northern auroral X-ray hot spot pulsing with a period of about 45 minutes on 18 December 2000 [45]. XMM-Newton and Chandra campaigns in March 2007 and May–June 2016 found the southern X-ray aurora pulsing every 11 minutes while the northern aurora varied independently of it [30]. The two poles are a pair whose pulses are not locked to each other.

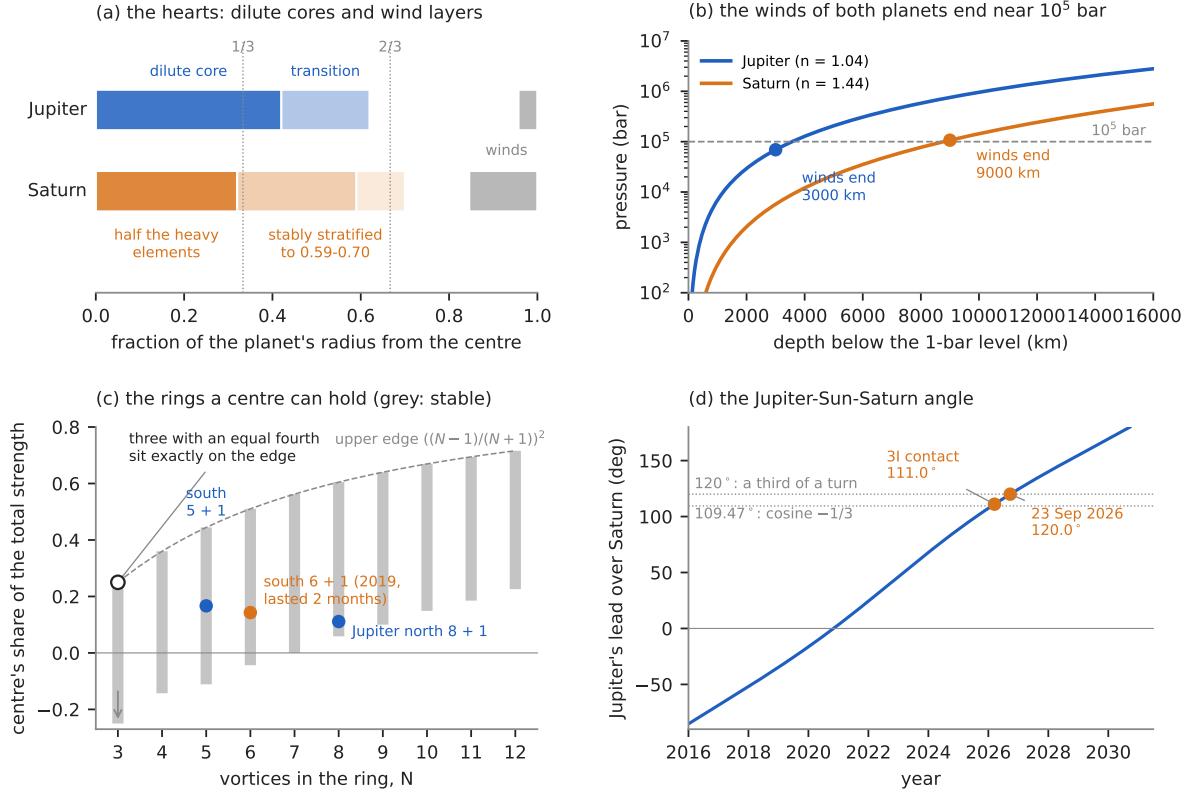


Figure 2: (a) The hearts by fraction of radius: Jupiter’s dilute core and transition layer [101], Saturn’s heavy-element core and stable region [92], and the wind layers (grey); dotted lines at $\frac{1}{3}$ and $\frac{2}{3}$. (b) Pressure against depth in the polytropes whose J_2/q_ω matches each planet; dots mark the published wind depths. (c) Grey: the central share $\kappa/(\kappa + N)$ for which a ring of N equal vortices around a centre of strength κ is spectrally stable; dashed, the upper edge $((N-1)/(N+1))^2$; dots, Jupiter’s polar patterns with the centre equal to the others. (d) Jupiter’s lead over Saturn in heliocentric longitude, with the 3I/ATLAS contact of Section 7.2.

5 The gas ladder

Each giant’s troposphere is a ladder of cloud decks. Water condenses deepest; above it ammonia and hydrogen sulfide combine into solid ammonium hydrosulfide; above that the survivor of that reaction condenses as ice; and in the two cold planets methane condenses on top. The equilibrium cloud-condensation model that computes this ladder goes back to Lewis and to Weidenschilling and Lewis [79, 153]; its modern form is that of Atreya and Wong [6]. We build it from stated thermodynamic data, validate it against the published decks, and show that its middle rung is an exact two-state pair whose balance decides what each planet shows above its clouds.

5.1 Inputs

Table 4 lists every external input. The deck bases depend on the 1-bar temperature, the abundances of the condensing species, the saturation vapour pressures and the heat capacity of the gas; the gravity enters only the conversion from pressure to altitude.

5.2 The model

(i) Composition. Deep mole fractions x_i of the trace species are given either directly (Juno, Molter et al., Fletcher et al., ice-giant CH_4) or as ratios r_i to H_2 (Galileo, protosolar multiples). Then

$$x_{\text{H}_2} = \frac{1 - \sum_{\text{given}} x_j}{1 + \text{He}/\text{H}_2 + \sum_i r_i}, \quad x_{\text{He}} = (\text{He}/\text{H}_2) x_{\text{H}_2}, \quad x_i = r_i x_{\text{H}_2}.$$

The nominal abundances are, for Jupiter, NH_3 362 ppmv (Juno), $\text{H}_2\text{S}/\text{H}_2 = 8.9 \times 10^{-5}$ (Galileo) and H_2O 2500 ppmv (Juno); for Saturn, NH_3 4×10^{-4} and S and O at $5 \times$ protosolar; for Uranus, the NH_3 and H_2S of Molter et al., CH_4 4% and O $30 \times$ protosolar; for Neptune, N $1 \times$, S $30 \times$ and O $30 \times$ protosolar, CH_4 4%. They give the mole fractions of Table 5. We write $\text{N} = x_{\text{NH}_3}$ and $\text{S} = x_{\text{H}_2\text{S}}$ for the deep values.

(ii) The heat capacity of hydrogen, and its ortho/para state. Take the rotational levels of the vibrational ground state, $E_J/hc = B_0 J(J+1) - D_e[J(J+1)]^2$ with $B_0 = B_e - \alpha_e/2 = 59.3219 \text{ cm}^{-1}$, which gives $E_1 = 118.455$, $E_2 = 354.236$, $E_3 = 705.080$ and $E_4 = 1167.60 \text{ cm}^{-1}$; we keep $J \leq 15$, where the levels still increase. The nuclear-spin weights are 1 for para (even J) and 3 for ortho (odd J), times $2J+1$. For levels with weights g_J and reduced energies $x_J = hcE_J/k_B T$, the rotational heat capacity is the variance $\text{Var}(x)$ under the weights $g_J e^{-x_J}$ (the second derivative of $\ln Z$ in $1/T$). Then, for normal H_2 , with ortho:para frozen at 3:1,

$$\frac{c_p}{R} = \frac{5}{2} + \frac{1}{4} \text{Var}_{\text{para}} + \frac{3}{4} \text{Var}_{\text{ortho}} + c_{\text{vib}},$$

and for equilibrium H_2

$$\frac{c_p}{R} = \frac{5}{2} + \text{Var}_{\text{all}} + c_{\text{vib}},$$

where the second includes the heat of ortho–para conversion and c_{vib} is the harmonic term for the fundamental interval $\omega_e - 2\omega_e x_e = 4158.541 \text{ cm}^{-1}$. Against the NIST fluid data at 0.1 bar and 40–400 K [78], the computed c_p agrees to 0.24% for normal H_2 (22.565 against 22.578 $\text{J mol}^{-1} \text{K}^{-1}$ at 100 K) and to 0.24% for para H_2 (27.007 against 27.015). No single value of c_p fits the tropospheres, which span 70–600 K (Table 6). The nominal model uses normal H_2 with $c_p(T)$ from these levels, He with $c_p/R = \frac{5}{2}$ and the polyatomic vapours as rigid non-linear rotors, $c_p/R = 4$.

(iii) Dry adiabat. For an ideal mixture with molar heat capacity $\bar{c}_p(T) = \sum_i x_i c_{p,i}(T)$ over the local gas,

$$\frac{d \ln T}{d \ln P} = \frac{R}{\bar{c}_p(T)}, \quad T(1 \text{ bar}) = T_{\text{NSSDC}},$$

integrated upward to 0.2 bar and downward to 300 bar (Jupiter, Saturn) or 3000 bar (Uranus, Neptune) by the fourth-order Runge–Kutta method in $\ln P$ with step 0.002. Condensable vapours above their decks are held at their local saturation values in $\bar{c}_p$. Anchored at the Galileo 1-bar temperature of 166.1 K, the Jupiter adiabat gives 426.65 K at 22 bar; the probe measured $427.7 \pm 1.5 \text{ K}$ (Seiff et al., as quoted in [50]).

input	value	source
T at 1 bar	Jupiter 165 K, Saturn 134 K, Uranus 76 K, Neptune 72 K	NSSDC [159, 161, 162, 160]
gravity at 1 bar	23.12, 8.96, 8.69, 11.00 m s ⁻² (equatorial)	NSSDC
T at 0.1 bar (context)	112, 84, 53, 55 K	NSSDC
bulk composition (context)	J: H ₂ 89.8%, He 10.2%; S: 96.3%, 3.25%; U: 82.5%, 15.2%, CH ₄ 2.3%; N: 80.0%, 19.0%, CH ₄ 1.5%	NSSDC
protosolar X/H	C 2.95×10^{-4} , N 7.41×10^{-5} , O 5.37×10^{-4} , S 1.45×10^{-5} (N/S = 5.110)	[2], photospheric +0.04 dex, as tabulated in [5]
He/H ₂	J 0.157; S 0.135 (alternative 0.124); U 0.180; N 0.241 (0.180 with N ₂)	[5]
Jupiter NH ₃	362 ± 33 ppmv (nominal); 351^{+22}_{-21} ppmv; Galileo probe N/H = $(3.32 \pm 1.27) \times 10^{-4}$ (566 $\pm$ 216 ppmv); depleted 100–250 ppmv at mid-latitudes	[80, 81, 166, 50]
Jupiter H ₂ S	S/H = $(4.45 \pm 1.05) \times 10^{-5}$ (H ₂ S/H ₂ = $(8.9 \pm 2.1) \times 10^{-5}$)	[166]
Jupiter H ₂ O	$2.5^{+2.2}_{-1.6} \times 10^3$ ppmv at 0–4°N; 490 ± 160 ppmv in the 5- μ m hot spot	[81, 166]
Jupiter CH ₄	C/H = 1.19×10^{-3}	[166], via [5]
Saturn CH ₄ , NH ₃	$(4.7 \pm 0.2) \times 10^{-3}$; $(4 \pm 1) \times 10^{-4}$ (observed to 3 bar)	[40, 5]
Saturn S, O	S: 1 $\times$ protosolar to S/H = 1.88×10^{-4} (13 $\times$, marked highly uncertain by the compilers), nominal 5 $\times$; O: 1–10 $\times$, nominal 5 $\times$	[5, 6]
Uranus NH ₃ , H ₂ S	deep $1.7^{+0.7}_{-0.4} \times 10^{-4}$ and $8.7^{+3.1}_{-1.5} \times 10^{-4}$, N/S = $0.20^{+0.08}_{-0.07}$; above the cloud 0.4–0.8 ppm H ₂ S at the 1.2–3 bar cloud top, below it $> (1.0\text{--}2.5) \times 10^{-5}$	[105, 61]
Neptune H ₂ S	1–3 ppm at the 2.5–3.5 bar cloud tops, below the cloud $> (0.4\text{--}1.3) \times 10^{-5}$; deep H ₂ S 10–30 $\times$ solar, bulk S/N $> 5\times$ solar	[62, 25]
Uranus, Neptune CH ₄	$\approx 4\%$ (equatorial) to $\approx 2\%$ (polar) near the 2–3 bar cloud tops	[63]
ice-giant H ₂ O	1, 10, 30, 80 $\times$ protosolar; nominal 30 $\times$	[5, 105]
NH ₄ SH equilibrium	$\log_{10}(P_{\text{NH}_3}P_{\text{H}_2\text{S}}/\text{atm}^2) = 14.82 - 4705/T$	[6], eq. (3)
H ₂ constants	$\omega_e = 4401.213$, $\omega_e x_e = 121.336$, $B_e = 60.8530$, $\alpha_e = 3.0622$, $D_e = 0.0471$ cm ⁻¹	[55], via [83]
H ₂ heat capacity (validation)	normal and para H ₂ , 0.1 bar, 40–400 K	[78], via [83]
H ₂ O vapour pressure	liquid: IAPWS SR1-86, $T_c = 647.096$ K, $p_c = 22.064$ MPa; ice Ih: IAPWS R14-08, $T_t = 273.16$ K, $p_t = 611.657$ Pa	[150, 149]
NH ₃	$T_{\text{fus}} = 194.95$ K, $P_{\text{triple}} = 0.06060$ bar, $\Delta_{\text{sub}}H = 31.2$ kJ mol ⁻¹ ; Antoine $3.18757 - 506.713/(T - 80.78)$ (164–239.6 K), $4.86886 - 1113.928/(T - 10.409)$ (239.6–371.5 K), $\log_{10} P/\text{bar}$	[83]
H ₂ S	$T_{\text{triple}} = 187.66$ K, $P_{\text{triple}} = 0.232$ bar, $T_{\text{boil}} = 212.87$ K; $\Delta_{\text{sub}}H = 22.5$ kJ mol ⁻¹ at 135 K, 25.4 at 175 K; Antoine $4.43681 - 829.439/(T - 25.412)$ (138.8–212.8 K), $4.52887 - 958.587/(T - 0.539)$ (212.8–349.5 K)	[83]
CH ₄	$T_{\text{triple}} = 90.67$ K, $P_{\text{triple}} = 0.1169$ bar; $\Delta_{\text{sub}}H = 9.7$ kJ mol ⁻¹ at 72 K; liquid Antoine $3.9895 - 443.028/(T - 0.49)$ (90.99–189.99 K)	[83]
constants	$R = 8.314462618$ J mol ⁻¹ K ⁻¹ , $hc/k_B = 1.438776877$ cm K	[19]

Table 4: Inputs of the gas ladder. The NIST WebBook compiles the NH₃, H₂S and CH₄ data from their primary sources, listed in Section 10.

(iv) Saturation vapour pressures. Water follows the IAPWS liquid equation above 273.16 K and the IAPWS ice-Ih sublimation equation below, with no condensation above 647.096 K [150, 149]. Solid NH₃, H₂S and CH₄ follow the Clausius–Clapeyron law through the

	H ₂	He	CH ₄	NH ₃	H ₂ S	H ₂ O	N/S	$\Delta = N - S$
Jupiter	0.8600	0.1350	2.047×10^{-3}	3.620×10^{-4}	7.654×10^{-5}	2.500×10^{-3}	4.730	$+2.855 \times 10^{-4}$
Saturn	0.8723	0.1178	4.700×10^{-3}	4.000×10^{-4}	1.265×10^{-4}	4.684×10^{-3}	3.162	$+2.735 \times 10^{-4}$
Uranus	0.7911	0.1424	0.0400	1.700×10^{-4}	8.700×10^{-4}	2.549×10^{-2}	0.1954	-7.000×10^{-4}
Neptune	0.7534	0.1816	0.0400	1.117×10^{-4}	6.555×10^{-4}	2.427×10^{-2}	0.1703	-5.438×10^{-4}

Table 5: Nominal deep mole fractions. $N/S = x_{\text{NH}_3}/x_{\text{H}_2\text{S}}$ at depth; protosolar N/S is 5.11.

T (K)	60	76	80	100	120	140	165	200	300	400	600
normal	2.519	2.571	2.590	2.714	2.857	2.993	3.135	3.280	3.470	3.511	3.529
equilibrium	4.359	3.853	3.747	3.386	3.232	3.191	3.217	3.302	3.471	3.511	3.529

$R/\bar{c}_p$ at 1 bar	normal H ₂	equilibrium H ₂	mean molar mass (g mol ⁻¹)
Jupiter	0.3278	0.3203	2.318
Saturn	0.3442	0.3209	2.317
Uranus	0.3884	0.2740	2.453
Neptune	0.3922	0.2713	2.458

Table 6: Top: c_p/R of normal and equilibrium H₂. Bottom: the adiabatic exponent of each mixture at 1 bar.

triple point with constant sublimation enthalpy, $p_{\text{sat}} = p_t \exp[-\frac{L}{R}(\frac{1}{T} - \frac{1}{T_t})]$, with $(T_t, p_t, L) = (194.95 \text{ K}, 0.06060 \text{ bar}, 31.2 \text{ kJ mol}^{-1})$, $(187.66, 0.232, 22.5)$ and $(90.67, 0.1169, 9.7)$; liquids above their triple points follow the Antoine equations of Table 4. As sensitivity variants we extrapolate Stull’s solid-range Antoine equations below their lower limits (the NH₃ form diverges near 81 K and is spliced to the Clausius–Clapeyron shape below 120 K, far from any deck base), use $L = 25.4 \text{ kJ mol}^{-1}$ for H₂S, and vary L of CH₄ by $\pm 5\%$. The adopted and Stull curves give, for NH₃, 1.89×10^{-4} against 7.37×10^{-5} bar at 164 K and 1.60×10^{-3} against 1.26×10^{-3} bar at 180 K; for H₂S, 6.83×10^{-5} (adopted), 4.65×10^{-5} (Stull) and 2.39×10^{-5} bar ($L = 25.4$) at 120 K, and 1.71×10^{-3} against 1.58×10^{-3} bar at 140 K.

(v) The NH₄SH equilibrium. $\text{NH}_3(\text{g}) + \text{H}_2\text{S}(\text{g}) \rightleftharpoons \text{NH}_4\text{SH}(\text{s})$ with $K(T) = P_{\text{NH}_3}P_{\text{H}_2\text{S}}$ in atm² and $\log_{10} K = 14.82 - 4705/T$ [6]; the NH₃–H₂S cloud chemistry goes back to Lewis [79]. The van ’t Hoff enthalpy is $L_{\text{rx}} = 4705 R \ln 10 = 90.08 \text{ kJ mol}^{-1}$, and $d \ln K / dT = 10834/T^2$; for example $K(200 \text{ K}) = 1.97 \times 10^{-9}$ and $K(250 \text{ K}) = 1.00 \times 10^{-4} \text{ atm}^2$.

(vi) The deck base as a balance point. For a simple condensable with vapour mole fraction x , define the log saturation ratio

$$\varepsilon_{\text{deck}}(P) = \ln \frac{xP}{p_{\text{sat}}(T(P))}.$$

Condensate is stable where $\varepsilon_{\text{deck}} > 0$, and the deck base is the deepest level where $\varepsilon_{\text{deck}} = 0$, crossing from negative below to positive above; we find it by root bracketing on the profile. For NH₄SH, $\varepsilon_{\text{NH}_4\text{SH}} = \ln(x_{\text{NH}_3}x_{\text{H}_2\text{S}}P_{\text{atm}}^2/K(T))$. For the ice deck of the surviving species (NH₃ or H₂S), x is the mole fraction left by the NH₄SH equilibrium at that level (Section 5.5). Each deck is thus a two-state pair, vapour and condensate, and its base is the balance point where the saturation ratio is 1, $\varepsilon_{\text{deck}} = 0$.

(vii) Moist pseudo-adiabat (sensitivity). Let $L_k = RT^2 d \ln p_{\text{sat},k} / dT$ be the latent heats of the saturated species. The first law for one mole of gas reads $\bar{c}_p dT - RT d \ln P = -\sum_k L_k dx_k - L_{\text{rx}} dx_{\text{H}_2\text{S}}$. For a saturated species $x_k P = p_{\text{sat},k}(T)$, so $dx_k = x_k(L_k dT / RT^2 -$

$d \ln P$). For NH_4SH , $x_{\text{NH}_3}x_{\text{H}_2\text{S}}P^2 = K$ and the two gases are removed 1:1, so $dx_{\text{NH}_3} = dx_{\text{H}_2\text{S}} = Y(10834 dT/T^2 - 2 d \ln P)$ with $Y = x_{\text{NH}_3}x_{\text{H}_2\text{S}}/(x_{\text{NH}_3} + x_{\text{H}_2\text{S}})$; this counts the reaction heat once per pair removed. Substituting and solving for $d \ln T / d \ln P$ gives

$$\frac{d \ln T}{d \ln P} = \frac{R}{\bar{c}_p} \frac{1 + (\sum_k L_k x_k + 2L_{\text{rx}}Y)/RT}{1 + (\sum_k L_k^2 x_k / R + 10834 L_{\text{rx}}Y)/(\bar{c}_p T^2)},$$

which is equation (7) of [6] written in $\ln P$.

(viii) **Altitude.** $dz = -(RT/\bar{m}g) d \ln P$, with $\bar{m}$ the mean molar mass and the NSSDC equatorial gravity held constant, measured from 1 bar (positive upward).

5.3 Validation against published models

We evaluated the model with the inputs of published cloud models (Table 7); “AG solar” is the Anders–Grevesse solar composition tabulated by Atreya and Wong, N/H 1.12×10^{-4} , S/H 1.62×10^{-5} , O/H 8.51×10^{-4} , C/H 3.62×10^{-4} [6]. NH_3 ice on Jupiter lies within 1.5% ($1 \times$ solar) and 2.0% ($3 \times$ solar) of Atreya and Wong. Against the rounded values of Li et al. (0.7, 2.2, 5 bar) the model lies within 4.7% (NH_3), 7.4% (NH_4SH) and 4.2% (water, dry) or 1.6% (water, moist). The moist water bases lie 4.0–7.1% shallower than the published moist values for Jupiter $1 \times$ and Saturn $1 \times$ and $5 \times$; the dry ones lie 8.6–21.4% shallower, because with the 1-bar anchor a dry profile is warmer at depth. Uranus’s NH_3 ice (published ≈ 10 bar) lies 4.0% shallower moist and 22% dry. The ice-giant deep water bases differ for three reasons: the published models include the aqueous $\text{NH}_3\text{--H}_2\text{O}$ solution and van der Waals corrections, while this model treats pure water, whose liquid base ($T > 273$ K) corresponds to the base of the solution cloud; latent heat is large at $30\text{--}80 \times$ O, so the dry and moist profiles diverge strongly below the NH_4SH deck (290 against 948 bar at $30 \times$); and the published models fix their profiles at different levels. The published kilobar-scale values lie between the dry and moist bases of this model.

case	CH_4 ice	NH_3 ice	H_2S ice	NH_4SH	H_2O	published
Jupiter $1 \times$ AG	–	0.709 / 0.709	–	2.113 / 2.117	5.211 / 5.436	NH_3 720 mb; water 5.7 bar [6]
Jupiter $3 \times$ AG	–	0.823 / 0.823	–	2.481 / 2.494	6.608 / 7.397	NH_3 840 mb; water 5–8 bar for $1\text{--}3 \times$ [6]
Jupiter $1 \times$ protosolar	–	0.667 / 0.667	–	2.037 / 2.041	4.788 / 4.922	$\approx 0.7, 2.2, 5$ bar [81]
Saturn $1 \times$ AG	–	1.435 / 1.442	–	4.326 / 4.354	11.19 / 11.71	water 12.6 bar; NH_4SH “5 bar region” [6]
Saturn $5 \times$ AG	–	1.812 / 1.854	–	5.557 / 5.735	16.52 / 20.17	water 21 bar [6]
Neptune $1 \times$ AG	0.720	9.62 / 9.66	–	29.16 / 29.34	85.6 / 89.0 (liquid base, 323 K)	CH_4 at ≈ 1 bar; water ice ≈ 40 bar; solution cloud $\approx 2 \times$ deeper [6]
Neptune $30 \times$ AG	1.332 / 1.693	–	–	–	290.0 / 947.9	solution cloud 370 bar [6]
Neptune $50 \times$ AG	1.481 / 2.253	–	–	–	383.8 / 2623	500 bar [6]
Uranus: C $45 \times$, N, O, S $1 \times$, 76 K	1.187 / 1.479	7.79 / 9.60	–	24.46 / 30.25	–	CH_4 1–1.4 bar; NH_3 ≈ 10 bar [5]
Uranus: C $80 \times$, S/N = 6, N 0.001–0.1 $\times$	1.351 / 2.233	–	1.54–2.55 / 2.40–4.02	–	–	H_2S ice in the 2–3 bar region, ~ 3 bar for S/N = 6 [5]
Uranus O $1 \times$, $80 \times$	–	–	–	–	67.7 / 114.9; 315.0 / 1732	~ 1 kilobar ($80 \times$) [5]

Table 7: Validation. Deck-base pressures in bar, dry/moist, against the published values.

5.4 The decks and the rungs

Table 8 gives the nominal deck bases on the dry adiabat of normal H_2 . On the nominal profiles the temperature is, for Jupiter, 130.59 K at 0.5 bar, 146.55 at 0.7, 170.22 at 1.1, 188.09 at 1.5, 205.98 at 2, 255.0 at 4 and 424.1 at 22 bar; for Saturn 104.73, 138.45, 153.68, 168.91 and 210.43 K at 0.5, 1.1, 1.5, 2 and 4 bar; for Uranus 78.86, 88.70, 98.68, 107.02, 114.23, 120.61 and 126.36 K at 1.1, 1.5, 2, 2.5, 3, 3.5 and 4 bar; for Neptune 74.74, 84.23, 93.82, 101.86, 108.84, 115.01 and 120.58 K at the same levels.

planet	deck	P (bar)	T (K)	depth below 1 bar (km)	phase at base
Jupiter	NH ₃ ice	0.7628	150.84	−6.6 (above)	solid
	NH ₄ SH	2.3804	217.47	25.6	solid
	H ₂ O	5.8319	285.74	60.2	liquid
Saturn	NH ₃ ice	1.5372	154.94	24.8	solid
	NH ₄ SH	5.1301	227.22	116.0	solid
	H ₂ O	14.6824	311.82	227.4	liquid
Uranus	CH ₄ ice	1.3643	85.60	9.2	solid
	H ₂ S ice	6.2634	147.16	67.2	solid
	NH ₄ SH	34.9800	253.31	179.0	solid
	H ₂ O	200.4875	425.80	364.7	liquid
Neptune	CH ₄ ice	1.6186	86.69	11.0	solid
	H ₂ S ice	6.9726	146.01	53.8	solid
	NH ₄ SH	37.5896	250.04	137.2	solid
	H ₂ O	224.5147	428.26	283.7	liquid

Table 8: Nominal deck bases (dry adiabat, normal H_2 , nominal abundances). The water decks have liquid bases, with ice above the 273.16 K level.

The rungs. Counted as the model gives them with nominal abundances, Jupiter and Saturn have 3 rungs (NH₃ ice, NH₄SH, H₂O) and Uranus and Neptune have 4 (CH₄ ice, H₂S ice, NH₄SH, H₂O). Jupiter and Saturn carry the triad of decks; the ice giants carry a fourth, methane, on top. The count is fixed by two facts. Each planet has exactly one ice deck from the titration survivor (Theorem 3), so the count is the same in all 179 abundance-range cases of Section 5.7, and only the identity of the survivor can change. CH₄ condenses only in the two cold planets, which have 76 and 72 K at 1 bar; it does not condense anywhere on the Jupiter and Saturn profiles from 0.2 bar downward.

5.5 The titration as an exact two-state pair

Theorem 3 (the survivor is the sign of the odd part). *Let N and S be the deep mole fractions of NH_3 and H_2S , and write $x_{NH_3} = ke^{\varepsilon/2}$, $x_{H_2S} = ke^{-\varepsilon/2}$ at every level where $NH_4SH(s)$ coexists with the gas, with $k^2 = K(T)/P_{\text{atm}}^2$. Then:*

1. the odd part $\Delta = x_{NH_3} - x_{H_2S} = N - S$ is conserved, and $\Delta = 2k \sinh(\varepsilon/2)$, so $\varepsilon = 2 \operatorname{arsinh}(\Delta/2k) = \ln(x_{NH_3}/x_{H_2S})$;
2. the NH_3 share of the vapour is the logistic $\phi_N = x_{NH_3}/(x_{NH_3} + x_{H_2S}) = 1/(1 + e^{-\varepsilon})$, with $\phi_N - \frac{1}{2} = \frac{1}{2} \tanh(\varepsilon/2)$;
3. at the deck base $k = \sqrt{NS}$ and $\varepsilon_{\text{base}} = \ln(N/S)$; above it $k \rightarrow 0$ and $\varepsilon \rightarrow \operatorname{sign}(\Delta) \cdot \infty$, so the surviving member is $\operatorname{sign}(N - S)$, with mole fraction $|\Delta|$;
4. at $N = S$, $\varepsilon = 0$ at every level, both gases fall together as $k \rightarrow 0$, and neither survives: this is the zero of the pair;
5. the deep signed survivor fraction is $(N - S)/(N + S) = \tanh(\frac{1}{2} \ln(N/S))$.

Proof. Where the solid coexists with the gas, $x_{\text{NH}_3}x_{\text{H}_2\text{S}}P_{\text{atm}}^2 = K(T)$, so $x_{\text{NH}_3}x_{\text{H}_2\text{S}} = k^2$, which the parametrisation satisfies identically; k is the common part (the geometric mean) and $\varepsilon = \ln(x_{\text{NH}_3}/x_{\text{H}_2\text{S}})$. The solid takes NH_3 and H_2S one for one, so Δ keeps its deep value $\text{N} - \text{S}$ from the deck base upward, and $\Delta = k(e^{\varepsilon/2} - e^{-\varepsilon/2}) = 2k \sinh(\varepsilon/2)$, which gives (1). (2) is $x_{\text{NH}_3}/(x_{\text{NH}_3} + x_{\text{H}_2\text{S}}) = e^{\varepsilon/2}/(e^{\varepsilon/2} + e^{-\varepsilon/2})$. At the base the gas still has its deep composition, so $k^2 = \text{NS}$ and $\varepsilon = \ln(\text{N}/\text{S})$. Above the base the temperature falls, $K \propto 10^{-4705/T}$ falls faster than P^2 , and $k \rightarrow 0$ at fixed Δ ; by (1) $|\varepsilon| \rightarrow \infty$ with the sign of Δ , so the minority member vanishes ($x = ke^{-|\varepsilon|/2} \rightarrow 0$) and the majority member tends to $|\Delta|$, which is (3). If $\Delta = 0$ then $\varepsilon = 0$ at every level and $x_{\text{NH}_3} = x_{\text{H}_2\text{S}} = k \rightarrow 0$, which is (4). (5) is (2) evaluated at the base, $2\phi_{\text{N}} - 1 = \tanh(\varepsilon_{\text{base}}/2)$. $\square$

The titration is the logistic of the chemistry companion [137], $p = 1/(1 + e^\varepsilon)$ for the H_2S share, with the difference $\delta = \phi_{\text{N}} - \frac{1}{2}$ carried by ε ; the common part k cancels from the share, and the observer above the deck reads only the sign of the odd part. On the nominal profiles (Table 9) ε at the base equals $\ln(\text{N}/\text{S})$ to all digits, and 40 K above the base it has reached ± 8 to ± 11 . At the base of the surviving ice deck the minority species has fallen to 2.6×10^{-13} (H_2S , Jupiter), 4.5×10^{-13} (H_2S , Saturn), 2.6×10^{-16} (NH_3 , Uranus) and 1.5×10^{-16} (NH_3 , Neptune).

planet	level	P (bar)	x_{NH_3}	$x_{\text{H}_2\text{S}}$	ε
Jupiter	base (217.5 K)	2.380	3.620×10^{-4}	7.654×10^{-5}	1.554
	1 K colder	2.345	3.502×10^{-4}	6.475×10^{-5}	1.688
	5 K colder	2.209	3.169×10^{-4}	3.144×10^{-5}	2.310
	20 K colder	1.749	2.866×10^{-4}	1.153×10^{-6}	5.516
	40 K colder	1.251	2.855×10^{-4}	4.669×10^{-9}	11.02
Saturn	base (227.2 K)	5.130	4.000×10^{-4}	1.265×10^{-4}	1.151
	5 K colder	4.772	3.335×10^{-4}	5.998×10^{-5}	1.716
	20 K colder	3.807	2.768×10^{-4}	3.329×10^{-6}	4.421
	40 K colder	2.756	2.735×10^{-4}	2.413×10^{-8}	9.336
Uranus	base (253.3 K)	34.98	1.700×10^{-4}	8.700×10^{-4}	-1.633
	5 K colder	32.75	9.023×10^{-5}	7.902×10^{-4}	-2.170
	20 K colder	26.69	9.160×10^{-6}	7.092×10^{-4}	-4.349
	40 K colder	19.95	2.137×10^{-7}	7.002×10^{-4}	-8.095
Neptune	base (250.0 K)	37.59	1.117×10^{-4}	6.555×10^{-4}	-1.770
	5 K colder	35.20	5.736×10^{-5}	6.012×10^{-4}	-2.349
	20 K colder	28.68	5.296×10^{-6}	5.491×10^{-4}	-4.641
	40 K colder	21.41	1.082×10^{-7}	5.439×10^{-4}	-8.523

Table 9: The titration along the nominal profiles: $\varepsilon = \ln(x_{\text{NH}_3}/x_{\text{H}_2\text{S}})$ runs from $\ln(\text{N}/\text{S})$ at the base towards $\text{sign}(\text{N} - \text{S}) \cdot \infty$.

The zero. On each planet’s nominal profile we varied N/S over 41 values from 0.05 to 20 at fixed $\text{N} + \text{S}$. The top ice deck is H_2S for every $\text{N}/\text{S} < 1$ and NH_3 for every $\text{N}/\text{S} > 1$, and at $\text{N}/\text{S} = 1.000$ neither ice forms, on all four planets. The grid points flanking the zero are 0.861 (H_2S) and 1.162 (NH_3); on the Uranus profile they give H_2S ice at 4.25 bar and NH_3 ice at 7.64 bar, on Jupiter’s H_2S ice at 0.314 bar and NH_3 ice at 0.579 bar (Fig. 3c).

The planets. With protosolar $\text{N}/\text{S} = 5.11$:

- *Jupiter:* $\text{N}/\text{S} = 4.730$ (range 3.48–13.41); NH_3 survives with 2.855×10^{-4} . With Juno’s mid-latitude depleted NH_3 of 100–250 ppmv, $\text{N}/\text{S} = 1.306$ –3.266, still above 1; the NH_3 ice base moves to 0.556 bar (100 ppmv) and 0.713 bar (250 ppmv), and NH_4SH to 2.167 and 2.316 bar.

- *Saturn*: $N/S = 3.162$ (range 0.909–19.86). The one corner below 1 combines the compilers’ uncertain $13\times S$ with the low NH_3 (3×10^{-4}), and gives an H_2S ice deck at 0.62 bar instead of NH_3 ice.
- *Uranus*: $N/S = 0.195$ (range 0.110–0.333); H_2S survives with 7.00×10^{-4} .
- *Neptune*: $N/S = 0.170$ (range 0.170–1.022); H_2S survives with 5.44×10^{-4} . The upper corner ($N\ 2\times$, $S\ 10\times$) gives $N/S = 1.022$ and an NH_3 ice deck at 6.0 bar; the H_2S observed above Neptune’s clouds [62] excludes it.

5.6 Why the ice giants show hydrogen sulfide

The observable above the NH_4SH deck is the sign of the odd part, $\text{sign}(N - S)$ (Theorem 3). For deep $N/S < 1$ the titration consumes all the NH_3 at the NH_4SH deck (Uranus 35.0 bar, Neptune 37.6 bar) and leaves $|N - S|$ of H_2S (7.0×10^{-4} and 5.4×10^{-4}), which condenses as the H_2S ice deck at 6.26 and 6.97 bar; above it NH_3 falls below 10^{-15} . Uranus’s $N/S = 0.20$ is measured [105]; Neptune’s $N/S < 1$ is required by the observed H_2S [62] and by the microwave spectra [25]. For Jupiter and Saturn ($N/S = 4.73, 3.16$) the same equation gives the opposite sign and an NH_3 ice top deck. The zero, $N/S = 1$, is where the observable changes, and the scan shows that it changes exactly there. The comparison with observation is as follows.

- *Jupiter*. NH_3 ice is the top deck (0.763 bar). Galileo imaging found a nearly ubiquitous cloud base at 750 ± 200 mb (Banfield et al., as quoted in [6]); spectrally identifiable NH_3 ice covers about 1% of the planet (Baines et al., as quoted in [6]); the Galileo probe nephelometer saw tenuous layers at 0.55, 1.3 and 1.6 bar in a dry hot spot, which Atreya and Wong reproduce with depleted abundances.
- *Saturn*. NH_3 ice is the top deck (1.537 bar). Spectral evidence for NH_3 ice in the Great Storm of 2010–2011 is reported by Sromovsky, Baines and Fry [141]. The adopted tropospheric NH_3 , $(4 \pm 1) \times 10^{-4}$, is the NH_3 that survives above NH_4SH , as the titration sign requires.
- *Uranus*. H_2S survives and condenses at 6.26 bar. H_2S gas is detected above the main cloud at 0.4–0.8 ppm, with cloud tops at 1.2–3 bar [61]; the model’s saturated H_2S at 2–2.5 bar is 0.26–1.8 ppm, which brackets the detection. The sub-cloud H_2S of 7.0×10^{-4} satisfies the observed lower limit $(1.0\text{--}2.5) \times 10^{-5}$. The NH_4SH base at 35.0 bar agrees with the ≈ 35 bar of Molter et al. [105], and the CH_4 base (1.36 bar; 1.16–1.36 over the range) lies in the 1–1.4 bar region of the Voyager cloud layer [5].
- *Neptune*. H_2S survives and condenses at 6.97 bar. H_2S is detected at 1–3 ppm at the 2.5–3.5 bar cloud tops [62]; the model’s saturated H_2S at 2.5–3 bar is 0.49–2.25 ppm. The sub-cloud 5.4×10^{-4} satisfies the lower limit $(0.4\text{--}1.3) \times 10^{-5}$. Irwin et al. conclude that the observed cloud base likely lies deeper than 4 bar, consistent with 6.97 bar, and the CH_4 base (1.62 bar at 4%, 1.38 bar at 2%) matches the 1.5 bar methane condensation level they adopt [62].

5.7 Sensitivities and abundance ranges

Table 10 gives the deck bases under each variant of the model. The ortho/para state of hydrogen is the dominant model uncertainty in the ice giants: equilibrium H_2 moves Uranus’s decks deeper by 16% (CH_4), 50% (H_2S), 52% (NH_4SH) and 67% (H_2O), and Neptune’s by 26%, 57%, 60% and 78%, while Jupiter’s shift is at most 1.2% and Saturn’s at most 4.8%. The reason is visible in Table 6: at 70–90 K equilibrium H_2 has $c_p/R \approx 3.8$ against 2.6 for normal H_2 , so its adiabat is far shallower, while above 300 K the two coincide. The moist treatment has a similar size in the ice giants. No variant changes which decks exist, their order, or the identity of the surviving ice.

The abundance ranges are the full factorial of the literature ranges: for Jupiter NH_3 330, 362, 566, 782 ppmv $\times H_2S/H_2$ 6.8, 8.9, 11.0×10^{-5} $\times H_2O$ 490, 900, 2500, 4700 ppmv (48

variant	Jupiter NH ₃ /NH ₄ SH/H ₂ O	Saturn NH ₃ /NH ₄ SH/H ₂ O	Uranus CH ₄ /H ₂ S/NH ₄ SH/H ₂ O	Neptune CH ₄ /H ₂ S/NH ₄ SH/H ₂ O
nominal (dry, normal H ₂)	0.763/2.380/5.832	1.537/5.130/14.68	1.364/6.263/34.98/200.5	1.619/6.973/37.59/224.5
equilibrium H ₂	0.756/2.405/5.901	1.577/5.356/15.39	1.586/9.395/53.32/335.8	2.046/10.98/60.03/399.0
classical	0.736/2.557/6.469	1.651/6.124/18.22	1.536/10.15/63.85/423.5	1.935/11.54/69.65/484.5
$c_p/R = 3.5$ for H ₂				
moist	0.763/2.392/6.251	1.549/5.209/16.64	2.315/10.57/59.27/603.6	3.050/12.97/69.83/739.4
pseudo-adiabat				
polyatomic	0.763/2.381/5.837	1.538/5.138/14.72	1.366/6.355/35.96/210.9	1.622/7.073/38.62/236.1
$c_p/R = 4.5$				
NH ₃ ice: Stull	0.830 /2.380/5.832	1.641 /5.130/14.68	unchanged	unchanged
H ₂ S ice: Stull	unchanged	unchanged	1.364/ 6.312 /34.98/200.5	1.619/ 7.032 /37.59/224.5
H ₂ S ice:	unchanged	unchanged	1.364/ 6.841 /34.98/200.5	1.619/ 7.626 /37.59/224.5
$L = 25.4$ kJ mol ⁻¹				
CH ₄ ice $L \mp 5\%$	–	–	1.351, 1.376 (others $\pm 0.1\%$)	1.607, 1.629
$T(1 \text{ bar}) - 2 \text{ K}$	0.796/2.484/6.101	1.616/5.396/15.51	1.485/6.796/37.92/221.1	1.769/7.592/40.89/248.7
$T(1 \text{ bar}) + 2 \text{ K}$	0.732/2.282/5.577	1.463/4.880/13.91	1.256/5.783/32.32/182.2	1.484/6.418/34.63/203.3
He/H ₂	0.136:	0.124:	0.184:	0.180:
alternative	0.762/2.393/5.883	1.537/5.152/14.82; 0.0337 (NSSDC): 1.538/5.356/16.14	1.364/6.260/34.91/199.3	1.620/7.099/38.97/244.5

Table 10: Sensitivities: deck-base pressures in bar at nominal abundances. Bold: the deck the variant acts on.

cases); for Saturn NH₃ 3, 4, $5 \times 10^{-4} \times S$ 1, 5, $13 \times \times O$ 1, 5, $10 \times$ (27); for Uranus NH₃ 1.3, 1.7, $2.4 \times 10^{-4} \times H_2S$ 7.2, 8.7, $11.8 \times 10^{-4} \times CH_4$ 2%, 4% $\times O$ 1, 10, 30, $80 \times$ (72); for Neptune N 1, $2 \times \times S$ 10, $30 \times \times CH_4$ 2%, 4% $\times O$ 1, 10, 30, $80 \times$ (32). Table 11 gives the resulting decks. Over the ranges the depths below 1 bar are: Jupiter NH₃ 3.5–7.2 km above 1 bar, NH₄SH 24.7–28.1 km, water 47.4–66.5 km; Saturn 13.7–29.9, 103.0–124.8 and 184.6–249.7 km; Uranus CH₄ 4.5–9.2, H₂S 63.7–77.2, NH₄SH 175.6–200.6 and water 239.6–459.4 km; Neptune 7.3–11.0, 43.3–58.1, 131.1–153.9 and 187.6–357.2 km.

planet	N/S	CH ₄ ice	NH ₃ ice	H ₂ S ice	NH ₄ SH	H ₂ O
Jupiter	3.48– 13.41	–	0.743–0.869 (149.5–157.6 K), 48/48	–	2.317–2.564 (215.7–222.5 K)	4.31–6.72 (260.8–298.1 K)
Saturn	0.909– 19.86	–	1.277–1.665 (145.6–159.1 K), 24/27	0.616–0.624 (113.0–113.5 K), 3/27	4.432–5.644 (217.2–234.0 K)	10.17–17.59 (279.5–329.0 K)
Uranus	0.110– 0.333	1.163–1.364 (80.6–85.6 K)	–	5.734–6.789 (143.5–151.2 K), 72/72	32.64–37.15 (249.8–258.0 K)	66.0–317.7 (308.9–487.2 K)
Neptune	0.170– 1.022	1.376–1.619 (81.5–86.7 K)	6.006–6.127 (139.4–139.8 K), 8/32	5.136–7.006 (132.0–146.3 K), 24/32	33.13–40.18 (242.3–255.2 K)	74.4–355.0 (310.2–490.5 K)

Table 11: Deck bases in bar over the abundance ranges (dry, normal H₂), with the number of cases in which each ice is the top titration deck.

5.8 What the ladder structures: water and Jupiter’s ammonia

Where water is hidden. In every planet water is the deepest rung, below the titration deck, and its depth is set by O/H and by the 1-bar temperature. For 1–80× protosolar O on the dry adiabat the water base lies at 4.3–6.7 bar in Jupiter (47–66 km below 1 bar), 10.2–17.6 bar in Saturn (185–250 km), 66–318 bar in Uranus (240–459 km) and 74–355 bar in Neptune

(188–357 km). Against O/H with the other abundances nominal: Jupiter 4.31 bar (260.8 K, ice base) at 490 ppmv (hot spot), 4.77 bar (268.9 K, ice) at 900 ppmv, 5.83 bar (285.7 K, liquid) at 2500 ppmv and 6.72 bar (298.1 K, liquid) at 4700 ppmv; Saturn 10.17, 14.68 and 17.59 bar (279.5, 311.8, 329.0 K) at 1, 5 and 10 $\times$; Uranus 67.9, 132.1, 200.5 and 317.6 bar (309.1, 376.5, 425.8, 487.2 K) at 1, 10, 30 and 80 $\times$; Neptune 76.6, 148.3, 224.5 and 355.0 bar (310.4, 378.4, 428.3, 490.5 K). The moist profile carries the ice-giant bases to 0.6–1.7 kbar at 30–80 $\times$, and Uranus’s 80 $\times$ case to 1732 bar. The ladder places the water rung and gives its dependence on O/H; O/H itself is the open quantity, and in the ice giants the water base is further moved by the ortho/para state (up to 78%) and by the non-ideality of the solution cloud.

Jupiter’s ammonia depletion. Juno finds NH₃ variable and depleted (100–250 ppmv at mid-latitudes) down to tens of bars [80, 50]. Equilibrium condensation keeps 362 ppmv below the NH₄SH deck and removes only the H₂S-equivalent 7.7×10^{-5} at 2.38 bar, leaving the odd part 2.855×10^{-4} , so the depletion is dynamical. Guillot and co-workers explain it by the mushball mechanism [49, 50]: water ice lofted by storms into the 1.1–1.5 bar, 173–188 K region absorbs ammonia into a liquid of about $\frac{1}{3}$ NH₃ and $\frac{2}{3}$ H₂O, and the resulting hail and downdrafts carry ammonia to 7–25 bar. The nominal adiabat gives 170.2 K at 1.1 bar and 188.1 K at 1.5 bar, which places that region between the NH₃ ice (0.763 bar) and NH₄SH (2.380 bar) rungs. The titration sign is unchanged by the depletion: with 100–250 ppmv NH₃ and the Galileo H₂S, N/S = 1.31–3.27 > 1, so NH₃ remains the survivor and the top ice.

5.9 One pair law for the sensor and the deck

The chemistry companion [137] derives, from mass action on a metal-oxide surface, that the ionosorbed-oxygen occupancy is an exact logistic in the log-odds $\varepsilon = \ln(\Phi_+/\Phi_-)$ of the electron-release and trapping flux coefficients, linear in $\beta_R \ln C_R - \beta_O \ln C_O$ under three stated approximations, and measures the resulting sensor ladder in the UCI urban air record [27, 26]: on the common CO axis the two-state law identifies half-points $C_{1/2} = 1.68, 4.61$ and 5.40 mg m^{-3} for three sensors and places a fourth at 7.49 mg m^{-3} , at the upper edge of the data, with slopes $\beta = 1.18$ – 1.45 . In both systems the observable is the occupancy of a two-state pair fixed by a mass-action balance between two opposing species; its log-odds is linear in the log-ratio of their activities, and its balance point $\varepsilon = 0$ decides which member is observed (Table 12, Fig. 3d).

	metal-oxide sensor [137]	NH ₄ SH deck (Theorem 3)
pair	a surface site holds the electron (O [−]) or releases it	the surviving vapour is NH ₃ or H ₂ S
balance	$\theta/(1 - \theta) = \Phi_-/\Phi_+$	$x_{\text{NH}_3}/x_{\text{H}_2\text{S}} = e^\varepsilon$ with $x_{\text{NH}_3}x_{\text{H}_2\text{S}} = K/P^2$
log-odds	$\varepsilon = \ln \Phi_+/\Phi_- =$ $\beta_R \ln C_R - \beta_O \ln C_O + \text{const}$	$\varepsilon = 2 \operatorname{arsinh}[\Delta P/(2\sqrt{K})]$, = ln(N/S) at the base
occupancy	$1 - \theta = 1/(1 + e^{-\varepsilon})$	$\phi_N = 1/(1 + e^{-\varepsilon})$
common part (cancels)	any factor common to $\Phi_\pm$	$k = \sqrt{K}/P$, the geometric mean
zero	$C = C_{1/2}$: release flux equals trapping flux	N = S: neither survives
what the observer reads	which member holds the electron	which gas survives the deck

Table 12: One two-state law, two systems.

In both, two opposing reactants compete for a shared partner and mass action converts their ratio into an occupancy: on the sensor, reducing and oxidizing gases compete for the surface oxygen and its trapped electron; in the giants, NH₃ and H₂S are consumed together by the

NH_4SH lattice, one for one. The two differ in one respect: the sensor is an open steady state whose log-ratio is set by the ambient gases, with effective orders β fitted at 1.18–1.45, while the deck is a closed equilibrium titration with exact stoichiometric order 1, whose log-ratio is driven to $\pm\infty$ by cooling. The air record measures the single-coordinate logistic over 8991 present hours of its 9357-hour record on four surfaces, whose binned readings collapse onto $1/(1 + e^{-\varepsilon})$ with $\varepsilon = \beta \ln(C/C_{1/2})$; the four giants apply the two-member form of the same law, with the zero at $N/S = 1$ separating Jupiter and Saturn from Uranus and Neptune. The survivor $\text{sign}(N - S)$ takes the three values $+1$ (NH_3), 0 (neither) and -1 (H_2S): the digits of balanced ternary [138], the triad $\{-s, 0, s\}$ read by the observer above the clouds.

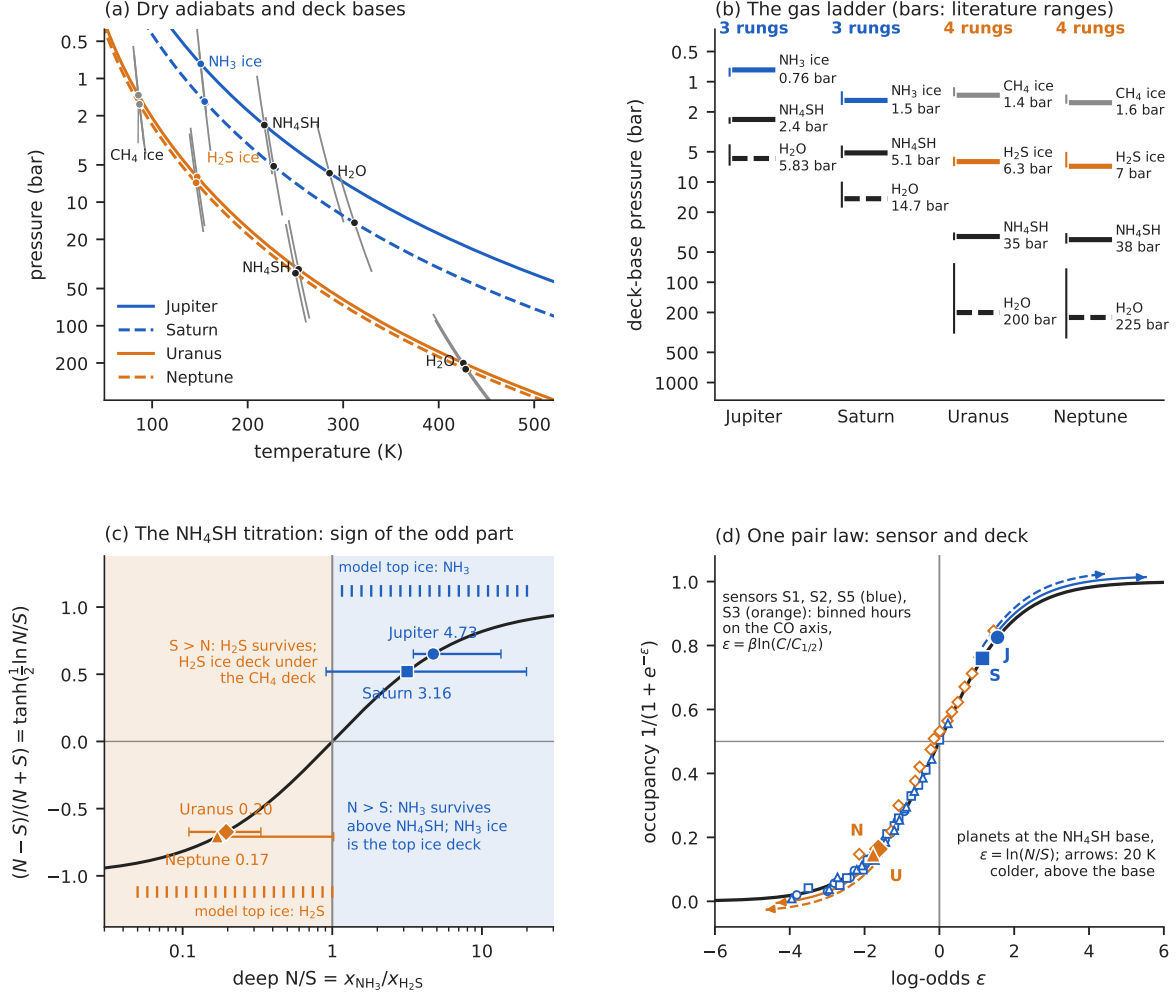


Figure 3: The gas ladder of the four giants from the equilibrium cloud-condensation model: dry adiabat of normal H_2 with He and CH_4 , anchored at the NSSDC 1-bar temperatures (165, 134, 76, 72 K). Blue: deep $N/S > 1$ (Jupiter solid, Saturn dashed); orange: $N/S < 1$ (Uranus solid, Neptune dashed). (a) Pressure–temperature profiles with each deck’s condensation curve (grey, for the planet’s nominal abundances) and its base (dots, labelled for Jupiter and Uranus). (b) The ladder of deck bases: horizontal bars are the nominal bases, vertical bars span the bases over the literature abundance ranges (179 model cases); the rung count is shown at the top, 3, 3, 4, 4. (c) The NH_4SH titration: the signed survivor fraction $(N - S)/(N + S) = \tanh(\frac{1}{2} \ln N/S)$ against deep N/S , zero at $N/S = 1$, with the planets at their nominal N/S and literature ranges. Tick rows: the model’s top ice deck across the N/S scan on the Uranus profile, identical on all four profiles. (d) One pair law, the logistic $1/(1 + e^{-\varepsilon})$. Open markers: binned hours of the metal-oxide sensors S1, S2, S5 (blue) and S3 (orange) of the chemistry companion [137] on the CO axis, with $\varepsilon = \beta \ln(C/C_{1/2})$ and occupancy $(S - S_{\text{lo}})/(S_{\text{hi}} - S_{\text{lo}})$. Filled markers: the four planets at their NH_4SH bases, where $\varepsilon = \ln(N/S)$ exactly; arrows follow ε to 20 K colder, above the base, towards $\text{sign}(N - S) \cdot \infty$.

6 The giants as a set

6.1 The classical order and the week

Order the seven classical bodies by decreasing period: Saturn, Jupiter, Mars, Sun, Venus, Mercury, Moon (the Chaldean order). Give each hour of the day to the next body in this order, and name each day for the body of its first hour, as Cassius Dio describes [17]. Consecutive days then advance by $24 \bmod 7 = 3$ places, which draws the star heptagram $\{7/3\}$ through the period order, and the days come out as Saturday, Sunday, Monday, Tuesday (Mars), Wednesday (Mercury), Thursday (Jupiter), Friday (Venus). The week is the period order of the bodies read in steps of three.

6.2 Moon systems: a locked triad and a fourth, and pairs

The sidereal periods of the Galilean moons are Io 1.769138, Europa 3.551181, Ganymede 7.154553 and Callisto 16.689017 days [109], so their mean motions are 203.488931, 101.374726, 50.317609 and 21.571073 deg/day. The inner three form two pairs, each near 2:1 (period ratios 2.00729 and 2.01470), and the two pairs are tied to each other:

$$n_{\text{Io}} - 2n_{\text{Eu}} = 0.739479, \quad n_{\text{Eu}} - 2n_{\text{Ga}} = 0.739508 \text{ deg/day.}$$

Their difference, the Laplace combination $n_{\text{Io}} - 3n_{\text{Eu}} + 2n_{\text{Ga}}$, is -0.000029 deg/day, 1.4×10^{-7} of Io's motion and below the 1.1×10^{-4} deg/day allowed by the six-decimal rounding of the periods. The Io–Europa conjunctions recur every 3.5255 days and the Europa–Ganymede conjunctions every 7.0509 days, exactly twice as long. The three moons are one locked triad; the Laplace resonance holds its angle at 180° , so a triple conjunction never occurs. Callisto stands outside the lock, at a period ratio of 2.33264 to Ganymede: Jupiter's large moons are a triad with a fourth body beside it (Fig. 4b). Saturn's moons [110] lock in pairs: Tethys/Mimas = 2.00314 (2:1, +0.157%), Dione/Enceladus = 1.99743 (2:1, -0.128%) and Hyperion/Titan = 1.33434 (4:3, +0.075%).

6.3 Jupiter and Saturn meet at thirds of a turn

The period ratio of Saturn to Jupiter is $P_S/P_J = 10759.22/4332.589 = 2.48332$ [159, 161], close to $\frac{5}{2}$. Their synodic period is 7253.45 days, or 19.859 years. In that time Jupiter moves 602.698° , so each conjunction falls $242.698^\circ = 0.67416$ of a turn beyond the previous one; at an exact 5:2 ratio the advance would be $\frac{5/2}{3/2} - 1 = \frac{2}{3}$ of a turn. Three conjunctions therefore close a triangle, the trigon, which turns by $3 \times 242.698^\circ - 720^\circ = 8.095^\circ$ every 59.58 years.

Proposition 4 (the trigon and the great inequality). *Let n_J, n_S be the mean motions in turns per unit time. Each corner of the trigon turns at the rate $\frac{1}{3}(5n_S - 2n_J)$, so the trigon advances one third of a turn in exactly one period $1/|2n_J - 5n_S|$ of the great inequality.*

Proof. Conjunctions recur every $S = 1/(n_J - n_S)$. In that time Jupiter moves $n_J S$ turns, and the conjunction point advances $n_J S$ turns, which is $\frac{5}{3} + \Delta$ with $\Delta = n_J S - \frac{5}{3}$ the excess over the exact 5:2 advance of $\frac{2}{3}$ of a turn modulo 1. Every third conjunction returns to the same corner, moved on by 3Δ turns after $3S$. The corner's rate is $\Delta/S = n_J - \frac{5}{3}(n_J - n_S) = \frac{1}{3}(5n_S - 2n_J)$, and $\frac{1}{3}$ of a turn takes $1/|5n_S - 2n_J|$. $\square$

With the NSSDC periods the great inequality has period 883 years, and $120^\circ/8.095^\circ \times 59.58$ years = 883 years. From the ephemeris we find the heliocentric conjunctions of 1802–2080 (Fig. 4a). Successive advances vary from 224° to 255° because both orbits are eccentric; their

mean is 242.1° . Every third conjunction returns to the same corner, moved on by 8.6° on average: 159° (1802), 169° (1861), 178° (1921), 187° (1981) and 196° (2040); 25° (1821) to 62° (2060); and 280° (1842) to 309° (2080). The 2020 heliocentric conjunction, on 2 November at 301.5° , preceded the geocentric great conjunction of 21 December 2020 near 300.5° .

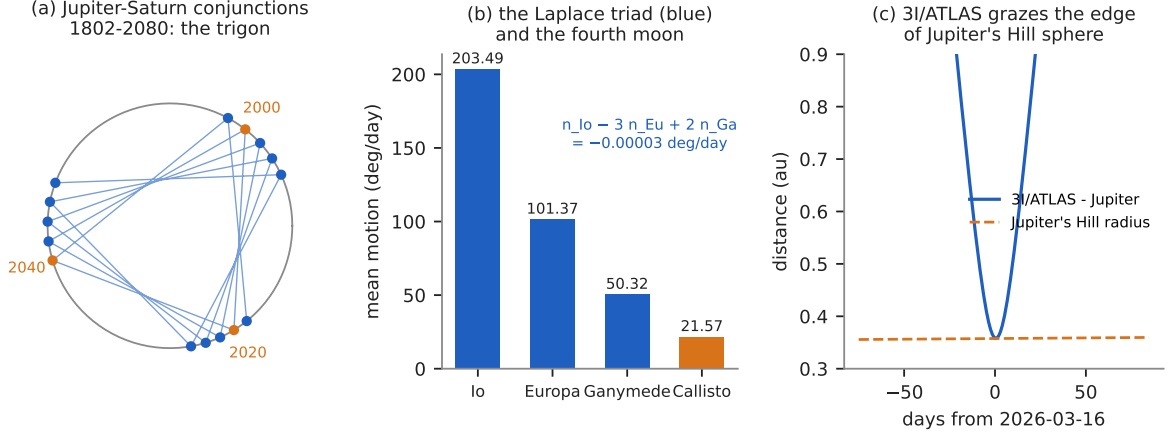


Figure 4: (a) Heliocentric Jupiter–Saturn conjunctions 1802–2080, joined in time order; each falls about two thirds of a turn after the last; those of 2000, 2020 and 2040 in orange. (b) Mean motions of the Galilean moons: the Laplace triad (blue) and Callisto (orange). (c) The distance of 3I/ATLAS from Jupiter (blue) and Jupiter’s Hill radius (orange), which meet at closest approach (Section 7.2).

6.4 The triangle is a triad with a centre

The equilateral configuration of three masses rotates rigidly. Gascheau and Routh [42, 121, 133] showed that it is linearly stable exactly when

$$(m_1 + m_2 + m_3)^2 > 27(m_1m_2 + m_2m_3 + m_3m_1).$$

The criterion forces a dominant centre: if m_1 is the largest mass and M the total, then $\sum_{i < j} m_i m_j \geq m_1(M - m_1)$, so $27x(1 - x) < 1$ with $x = m_1/M \geq \frac{1}{3}$, hence $m_1 > (1 - \mu_G)M = 0.9614791 M$, where $\mu_G = \frac{1}{2}(1 - \sqrt{23/27}) = 0.0385209$ is Routh’s critical value [135, 98]. For a planet and a test body the criterion reads $27\mu(1 - \mu) < 1$, that is $\mu < \mu_G$. The computed spectra agree with the criterion for 2000 values of μ and for 2000 random mass triples spanning six decades (870 of them stable), with real parts below 1.05×10^{-7} of the rotation rate on the stable side and above 1.34×10^{-2} on the unstable side. Three equal masses are unstable, with growth rate $0.7071 = 1/\sqrt{2}$ of the rotation rate, the equal-mass Lagrange triangle of the physics companion [135]: a triangle stands only around a dominant centre.

Proposition 5 (on the axis, and off it in mirrored fours). *At L_4 of the restricted problem the characteristic polynomial is $\lambda^4 + \lambda^2 + \frac{27}{4}\mu(1 - \mu) = 0$. For $\mu < \mu_G$ its roots are two pairs $\pm i\omega_1$, $\pm i\omega_2$ on the imaginary axis with $\omega_1^2 + \omega_2^2 = 1$ and $\omega_1^2\omega_2^2 = \frac{27}{4}\mu(1 - \mu)$; they meet at $\omega = 1/\sqrt{2}$ exactly at $\mu = \mu_G$, and for $\mu > \mu_G$ they form the quadruple $\pm a \pm ib$ off the axis.*

Proof. At L_4 , $U_{xx} = \frac{3}{4}$, $U_{yy} = \frac{9}{4}$ and $U_{xy} = \frac{3\sqrt{3}}{4}(1 - 2\mu)$ [107]. The linearised equations $\ddot{x} - 2\dot{y} = U_{xx}x + U_{xy}y$, $\ddot{y} + 2\dot{x} = U_{xy}x + U_{yy}y$ give $\lambda^4 + (4 - U_{xx} - U_{yy})\lambda^2 + U_{xx}U_{yy} - U_{xy}^2 = 0$, and $U_{xx}U_{yy} - U_{xy}^2 = \frac{27}{16}[1 - (1 - 2\mu)^2] = \frac{27}{4}\mu(1 - \mu)$. Hence $\lambda^2 = \frac{1}{2}[-1 \pm \sqrt{1 - 27\mu(1 - \mu)}]$, both negative for $27\mu(1 - \mu) < 1$, equal to $-\frac{1}{2}$ at equality, and complex conjugate beyond. $\square$

The eigenvalues of a real linear Hamiltonian system come in fours, $\{\lambda, -\lambda, \bar{\lambda}, -\bar{\lambda}\}$: the spectrum is invariant under the Klein four-group V generated by $\lambda \mapsto -\lambda$ (the Hamiltonian

symmetry) and $\lambda \mapsto \bar{\lambda}$ (reality). On the fixed line of $\lambda \mapsto -\bar{\lambda}$, the imaginary axis, an orbit of V collapses to a pair $\{\lambda, -\lambda\}$. The Krein signature of a pair on the axis is the sign of the quadratic Hamiltonian on its eigenspace, the sign of the mode's energy. Krein's theorem states that under a continuous Hamiltonian perturbation a pair of definite signature stays on the axis until it collides with a pair of opposite signature (or at zero) [98]. Below μ_G the two pairs at L_4 have opposite signatures on all 155 grid values: the slow libration carries negative energy (at $\mu = 10^{-6}$, $\omega = 0.002598$, signature -1) and the fast epicycle positive ($\omega = 0.999997$, $+1$). They meet at $1/\sqrt{2}$ at μ_G , and at $\mu_G(1 + 10^{-4})$ the four are $\pm 0.00346 \pm 0.70712i$ (Fig. 1b). With $\lambda = s - \frac{1}{2}$ the zeros of Riemann's ξ are invariant under the same group: $\xi(s) = \xi(1-s)$ is $\lambda \mapsto -\lambda$ and $\xi(\bar{s}) = \overline{\xi(s)}$ is $\lambda \mapsto \bar{\lambda}$; the fixed line of $\lambda \mapsto -\bar{\lambda}$ is the critical line, and the Riemann Hypothesis states that every orbit of V on the zero set is a pair on that line [134]. The Lee–Yang zeros of a real partition function that is symmetric under reversal of the field carry the same group in the variable $\lambda = \frac{1}{2} \ln z$: $z \mapsto 1/z$ is $\lambda \mapsto -\lambda$, $z \mapsto \bar{z}$ is $\lambda \mapsto \bar{\lambda}$, and the fixed line is the unit circle; there positive coupling keeps every zero on the circle, and a negative coupling sends zeros off it in mirrored pairs $z, 1/\bar{z}$ [136]. In all three settings one mechanism holds. A simple zero $iy \neq 0$ on the fixed line is an orbit of two, $\{iy, -iy\}$, and under a small perturbation that respects V it stays on the line: off the line its orbit would put two zeros (λ and $-\bar{\lambda}$) near iy , where the perturbed function has only one. Zeros leave the line only where two of them meet, and away from $\lambda = 0$ they leave as a quadruple $\{\lambda, -\lambda, \bar{\lambda}, -\bar{\lambda}\}$. Krein's theorem sharpens this for Hamiltonian spectra: a meeting sends two pairs off the axis only if their signatures are opposite. Below μ_G the two frequencies stand in the ratio 2:1 at $\mu = 0.0242939$ and 3:1 at $\mu = 0.0135160$, both inside the linearly stable range.

Trojans at all four. All four giants satisfy the criterion (the largest $27\mu(1-\mu)$ is Jupiter's, 2.57×10^{-2}), and Trojans are now known at all four: Jupiter's long-known clouds; Neptune's 35 by the 2023 count, 28 at L_4 and 7 at L_5 [155]; Uranus's two transient ones, 2011 QF₉₉ and 2014 YX₄₉ [23]; and Saturn's one, 2019 UO₁₄, confirmed in 2024, captured about 2000 years ago and leaving in about 1000 [57]. For primordial Trojans in the present planetary configuration, Nesvorný and Dones found Saturn's and Uranus's depleted about 100-fold over the age of the Solar System, while Neptune may retain about half [111]. Since the three-body triangles are linearly stable, the depletion comes from the rest of the planetary system, which is to each triangle a fourth body. The libration periods at L_4 are $P_{\text{orb}}/\omega_{\text{slow}}$, 147.4, 670.2, 4893 and 8837 years for the four giants (Table 3).

6.5 Four is three and a zero

To first order in the masses the inclinations of the four orbits evolve under the Laplace–Lagrange equations $\dot{\mathbf{p}} = B\mathbf{q}$, $\dot{\mathbf{q}} = -B\mathbf{p}$, with $p_j = I_j \sin \Omega_j$, $q_j = I_j \cos \Omega_j$ and the 4×4 matrix

$$B_{jk} = \frac{n_j}{4} \frac{m_k}{M_\odot + m_j} \alpha_{jk} \bar{\alpha}_{jk} b_{3/2}^{(1)}(\alpha_{jk}) \quad (j \neq k), \quad B_{jj} = - \sum_{k \neq j} B_{jk},$$

where $b_{3/2}^{(1)}$ is a Laplace coefficient [107].

Proposition 6 (the zero mode). *B has the eigenvalue 0 with eigenvector $(1, 1, 1, 1)$: a common tilt of all four orbits, which is a rotation of the reference plane, does not evolve. Its invariant plane is the invariable plane, perpendicular to the total orbital angular momentum.*

Proof. Every row of B sums to zero by construction. A common tilt changes all (p_j, q_j) by the same vector, which the equations leave fixed; the plane so defined is fixed by the conserved total angular momentum. $\square$

With the NSSDC masses and semimajor axes and the Laplace coefficients by quadrature, the eigenvalues of B are -25.5183 , -2.9276 , -0.6698 and 0 arcsec/yr, the zero to 6×10^{-17} of the largest. The three live modes have periods of 50,787, 442,677 and 1,935,044 years. The zero mode is occupied by no body; it is fixed by the total angular momentum, of which Jupiter carries 0.6158, Saturn 0.2501, Neptune 0.0800 and Uranus 0.0541. The eccentricities have four live modes, 0.6254, 2.7168, 3.6869 and 22.0865 arcsec/yr, and no zero.

6.6 No barrier between the four

Proposition 7 (tangent orbits lie below the throats). *An orbit coplanar with two circular planetary orbits, with perihelion on the inner and aphelion on the outer, has Tisserand parameter $\mathcal{T} = 1 - e + 2\sqrt{1+e}$ with respect to the inner planet and $\mathcal{T} = 1 + e + 2\sqrt{1-e}$ with respect to the outer, and both are below 3 for every $e > 0$.*

Proof. With respect to a planet at a_p , $\mathcal{T} = a_p/a + 2\sqrt{(a/a_p)(1-e^2)}$. Perihelion at a_p means $a = a_p/(1-e)$, which gives $\mathcal{T} = 1 - e + 2\sqrt{1+e}$; aphelion at a_p means $a = a_p/(1+e)$, which gives $1 + e + 2\sqrt{1-e}$. Both equal 3 at $e = 0$ and have derivatives $-1 + (1+e)^{-1/2} < 0$ and $1 - (1-e)^{-1/2} < 0$. $\square$

Since $\mathcal{T} \approx C$ for a small body and $C(L_1), C(L_2) > 3$ (Table 3), every tangent orbit is energetically open to both throats of both planets. For Jupiter–Saturn, Saturn–Uranus and Uranus–Neptune, $\mathcal{T} = 2.981/2.974$, $2.976/2.966$ and $2.989/2.986$, below even $C(L_4)$: at these energies the zero-velocity surfaces place no barrier between the four realms. The tangent orbits take 10.04, 27.23 and 61.28 years per hop, and one tangent orbit from Jupiter to Neptune takes 37.21 years; the arches of chaos do it in under a decade [145].

6.7 Two thirds of the horizon is filled at three giants

The Hill radius $r_H = a(m/3M_\odot)^{1/3}$ bounds each giant’s family of moons. We read the moon tables: all 29 moons of Uranus and 16 of Neptune; for Saturn the 210 Norse-group moons with eccentricities and 35 Gallic and Inuit moons with semimajor axes only; for Jupiter 13 outer moons with eccentricities and 21 with semimajor axes only [155]. Every listed moon lies inside its Hill sphere.

The largest apocentres $a(1+e)$, in units of r_H , are 0.6326 for Jupiter (S/2017 J 14), 0.6255 for Saturn (S/2023 S 5), 0.4070 for Uranus (Ferdinand) and 0.6543 for Neptune (S/2021 N 1; Neso 0.6233) (Fig. 1c). Three of the four lie within 10% of two thirds, in 0.600 – $0.733 r_H$, with mean 0.6374. Nine of Jupiter’s twelve largest orbits are listed without eccentricities; those moons would carry the reach beyond $0.733 r_H$ only with $e > 0.638$, and Saturn’s only with $e > 1$. The reach agrees with the known dynamics of irregular satellites: planar retrograde orbits are long-lived to $a \approx 0.7 r_H$, prograde ones only to $0.4 r_H$ [112], and the moons with the largest apocentres at all four giants are retrograde.

The exception, and a prediction. Uranus’s moons stop at 0.41 of its Hill radius; its retrograde moon with the largest apocentre has $a/r_H = 0.29$, against 0.44, 0.39 and 0.44 for the other three giants. We predict that Uranus has undiscovered retrograde irregular moons with apocentres of 0.60 – $0.73 r_H$, 42–51 million km from the planet.

7 The visitors

7.1 Three interstellar objects on every measured axis

We compare the three known interstellar objects on every axis with published measurements. Orbital elements are JPL solutions: 1I from the Small-Body Database (solution 16) [68], 2I and 3I as quoted from JPL in [156, 157]. The other quantities come from the sources cited in each table; where studies disagree we quote the range. Directions, deflections, velocities relative to the local standard of rest (LSR) and traceback separations are computed from these inputs, with the solar motion $(U, V, W)_\odot = (11.1, 12.24, 7.25)$ km/s [129].

	1I/Oumuamua	2I/Borisov	3I/ATLAS
designation	A/2017 U1	C/2019 Q4	C/2025 N1
perihelion (TDB)	2017 Sep 9.51	2019 Dec 8.56	2025 Oct 29.51
q_p (au)	0.25591	2.0065	1.3564
e	1.20113	3.3565	6.1414
i	122.74°	44.05°	175.12°
v_∞ (km/s)	26.32	32.28	57.94
incoming RA, Dec	279.55°, +33.90° (Lyra)	32.81°, +59.46° (Cassiopeia)	295.05°, −19.07° (Sagittarius)
outgoing RA, Dec	357.83°, +24.70°	275.44°, −51.99°	95.18°, +19.76°
deflection	112.7°	34.7°	18.7°
$2 \arcsin(1/e)$			

Table 13: Direction and orbit. q_p is the perihelion distance. The incoming directions of 1I and 3I agree with the published radiant to 0.1° [64, 24]; that of 2I is computed from its elements. v_∞ from [89, 56, 24].

Direction. We asked whether the three incoming directions share a common point: whether they cluster (circumradius), lie on one great circle ($|\det[\mathbf{r}_1 \mathbf{r}_2 \mathbf{r}_3]|$), have orbit planes that share one line ($|\det|$ of the poles), or have a circumcentre near the solar apex, the Galactic Centre, the anticentre or the Galactic pole. Each statistic was compared with an isotropic population and a kinematic population (10^6 draws each). The kinematic population draws stellar velocities from a Gaussian ellipsoid with dispersions (35, 25, 18) km/s about the LSR, weighted by the encounter rate $v(1 + (42.1 \text{ km/s}/v)^2)$ with gravitational focusing at 1 au. The three directions share no point: none of the eight statistics departs from either population. The circumradius, 57.2°, lies below the population medians 69.6° and 66.1° with $p = 0.27$ (isotropic) and 0.35 (kinematic); the great circle gives $p = 0.73/0.78$, the common orbital line 0.76/0.77 and the reference points 0.87/0.89 (with the antipode rule the Galactic Centre and anticentre are one statistic); and the smallest enclosing circle, of radius 55.1°, gives $p = 0.32$. The directions carry known physics instead: 1I came from 15° from the solar apex, as expected for a slow object, and 3I from 28° from the Galactic Centre.

Origin. Members of one star, cluster or moving group differ in velocity by a few km/s. The three objects differ by up to 75.5 km/s relative to the LSR (34.6 for 1I–2I, 48.0 for 1I–3I, 75.5 for 2I–3I). Triples drawn from the kinematic population give a median largest difference of 75.4 km/s ($p = 0.50$), and a difference below 10 km/s occurs in 5×10^{-5} of them. On straight-line paths the three were 35–77 pc apart one million years ago and 354–772 pc apart ten million years ago, so they share no recent point of release: the three interstellar objects share neither a direction nor a kinematic origin. Speed (26 → 32 → 58 km/s), kinematic age (1 → 5 → 7 Gyr) and vertical excursion (0.016 → 0.12 → 0.48 kpc) rise together; the ages and excursions

	1I	2I	3I
heliocentric (U, V, W) (km/s)	(−11.46, −22.40, −7.75) [89]	(22.0, −23.6, 1.1) [142]	(−51.23, −19.46, 18.93) [24]
relative to LSR (km/s)	(−0.4, −10.2, −0.5); 10.2	(33.1, −11.4, 8.4); 36.0	(−40.1, −7.2, 26.2); 48.5
$z_{\max}$ (kpc) [71]	0.016	0.121	0.481
kinematic age (Gyr) [142]	$1.2^{+4.3}_{-1.0}$	$4.8^{+4.7}_{-2.8}$	$6.9^{+4.1}_{-3.3}$
population	young thin disk; no parent [89]	thin disk; Ross 573 fly-by, not a parent [8]	thick-disk-like, metal-poor host favoured [54, 71]

Table 14: Origin and kinematics.

are inferred from the velocities, so this is one fact, the age–velocity relation of the Galactic disk, by which faster objects are assigned to older, dynamically hotter parts of the disk [54, 142].

	1I	2I	3I
discovery	2017 Oct 19, Pan-STARRS1, Haleakalā [95]	2019 Aug 30, G. Borisov, Crimea [102]	2025 Jul 1, ATLAS Chile [103]
relative to perihelion	40 d after	100 d before	120 d before (TESS images from May [36])

Table 15: Time.

Time. Each object was found within about four months of perihelion, when it was brightest and nearest.

	1I	2I	3I
absolute magnitude	$H_V = 22.4 \pm 0.04$ [64]	nucleus hidden by coma	nucleus $H_V = 17.1 \pm 0.4$ [58]
radius	~ 0.1 km (albedo 0.04) [95]; Spitzer: < 49 –220 m [146]	0.2–0.5 km [69]	1.3 ± 0.2 km [58]; 0.4–1.2 km from non-grav. [144]
slope (%/100 nm)	9–23 (studies differ) [64]	10.6 ± 1.4 [56]	17.1 ± 0.2 [130]; 18.3 ± 0.9 [24]
class	D-type-like	like Solar System comets	slightly redder than D-type; TNO/Centaur-like
polarisation	not measured	high positive, like Hale–Bopp [7]	deep negative branch, -2.7% at 7° , inversion 17° [47]

Table 16: Brightness, size, colour and polarisation.

Brightness, colour, rotation. All three are moderately red and none is ultrared; nucleus sizes span roughly a factor of ten, with large uncertainties for the two active objects (Table 16). 3I’s negative polarisation branch is unprecedented [47]. 1I rotated in 8.67 ± 0.34 h [64] or 8.1 h [13], tumbling, with a lightcurve amplitude of 1.5–2.5 mag and axis ratio at least 6:1; 2I has no secure period; 3I rotates in 16.16 ± 0.01 h [124] or 16.79 ± 0.23 h [24], with amplitude 0.2–0.3 mag; the Hubble Space Telescope images were too sparsely sampled to fix a period [58]. The ratio $16.16/8.1 = 1.995$ lies within 0.3% of 2, the factor by which lightcurve periods are commonly

ambiguous. The free-floating planetary-mass object PSO J318.5–22 rotates in 8.6 ± 0.1 h [11] and lies 22° from the direction 3I came from, and 68° and 102° from those of 1I and 2I.

species	1I	2I	3I	Solar System comets
activity	none; dust $< 1.7 \times 10^{-3}$ kg/s [64]	coma at discovery; active by 7.8 au [167]	coma at discovery; CN from ≈ 3 au [120]	–
CO (CO/H ₂ O)	rate $< 9 \times 10^{23}$ s ^{−1} , no H ₂ O seen [146]	0.35–1.05 [21]; ≥ 1.7 [12]	1.65 ± 0.09 at 3.32 au [22]	median 0.03 [52]
CO ₂ (CO ₂ /H ₂ O)	rate $< 9 \times 10^{22}$ s ^{−1} [146]	not measured	7.6 ± 0.3 at 3.32 au [22]; 0.3–2.1 after perihelion [132]	median 0.12 [52]
C ₂	–	depleted (C ₂ /CN 0.1–0.3) [28]	strongly depleted [116]	mostly not depleted
nickel, iron	–	Ni and Fe; $\log(\text{Ni}/\text{Fe}) = 0.21 \pm 0.18$ [51, 114]	Ni without Fe before 2.64 au [120]; $\log(\text{Ni}/\text{Fe}) \approx 0.9 \rightarrow 0.3$ [59]	-0.06 ± 0.31 [90]
isotopes	–	–	$^{12}\text{C}/^{13}\text{C} \approx 151$, $^{14}\text{N}/^{15}\text{N}$ ≈ 363 , large errors [115]	≈ 90 , ≈ 150 [115]

Table 17: Activity and composition.

Activity, composition and acceleration. 2I and 3I share comae rich in carbon oxides relative to water (carbon monoxide for 2I, carbon dioxide and monoxide for 3I), far above Solar System medians; depleted C₂; and gas-phase nickel (Table 17). These point to formation or storage at low temperature, far from their stars. Nickel near 2 au is also seen in Solar System comets [90]; what is distinctive in 3I is nickel (Ni I [120]) *before* iron, with $\log(\text{Ni}/\text{Fe})$ falling from ≈ 0.9 towards Solar System values as it approached the Sun, which indicates a nickel carrier that releases at lower temperature [59]. 1I showed no gas or dust at deep limits, so its composition is unmeasured, and the proposed drivers of its acceleration (hydrogen from irradiated water ice, a nitrogen-ice fragment, radiation pressure on a thin body) remain to be decided [10, 65]. All three show non-gravitational acceleration: the radial parameter at 1 au is $(4.92 \pm 0.16) \times 10^{-6}$ m s^{−2} for 1I [99], $\approx 5 \times 10^{-8}$ m s^{−2} for 2I [56] and $\approx 1 \times 10^{-6}$ m s^{−2} for 3I [140]; the distance scalings differ, so the ordering is approximate. For 2I and 3I the acceleration matches their measured outgassing [56, 144]; for 1I no outgassing was seen.

Kin, not siblings. The answer on every axis is the same (Table 18). The three are samples of one population, the small bodies ejected from planetary systems throughout the Galactic disk, not three pieces of one source. They share what that population shares: they are red and small, and the two whose gas could be measured are rich in carbon oxides and carry nickel. They share no star, direction or date. What all three orbits have in common is the Sun as their focus, placed there by detection: interstellar objects are seen only when they pass close to us.

7.2 At the giants: 3I/ATLAS touched Jupiter’s Hill sphere

We propagated each object on its two-body hyperbola and found its closest approach to Mars, Jupiter and Saturn within ten years of perihelion (Table 19). As validation, our 3I–Jupiter approach is 0.358011 au at 2026-03-16 11:33 UT, against JPL’s 0.358330 au at 12:22 UT [157]; our 3I–Mars approach is 0.1936 au on 2025-10-03 and our 3I–Earth approach 1.798 au on 2025-12-19, both matching the published values.

axis	shared	not shared
direction	– (no common point)	Lyra, Cassiopeia, Sagittarius; spread typical of random directions
origin, time	Galactic disk	separate sources; inferred ages 1, 5, 7 Gyr
dates	found near perihelion (selection)	discovered in 2017, 2019 and 2025
size, brightness	small bodies	factor ~ 10 in radius
colour	moderately red, not ultrared	3I redder than 2I
polarisation	–	2I strongly positive; 3I deeply negative
rotation	–	1I tumbling, very elongated; 3I small amplitude (diluted by coma)
activity	2I and 3I active	1I inert
composition	2I and 3I: carbon-oxide-rich, C ₂ -depleted, nickel	3I: nickel before iron, CO ₂ -dominated
dynamics	non-gravitational acceleration in all three	strength and cause differ

Table 18: Synthesis: kin, not siblings.

object	planet	closest (au), date	r_H (au)	d/r_H
1I	Jupiter	4.8069, 2017-07-30	0.3720	12.9
1I	Saturn	8.1799, 2017-01-15	0.4599	17.8
2I	Jupiter	5.1385, 2020-05-25	0.3533	14.5
2I	Saturn	8.8901, 2020-10-27	0.4574	19.4
3I	Mars	0.1936, 2025-10-03	0.0073	26.4
3I	Jupiter	0.3580, 2026-03-16	0.3576	1.001
3I	Saturn	9.9791, 2025-05-28	0.4389	22.7

Table 19: Closest approaches and the planet’s Hill radius $r_H = r(m/3M_\odot)^{1/3}$ at its distance r from the Sun at that moment. 1I and 2I passed Mars at 190 and 225 Hill radii.

The horizon touched. Of the three interstellar objects, only 3I came near a giant planet. At its closest approach Jupiter was 5.238 au from the Sun, and its Hill radius, the distance of L_1 and L_2 where Jupiter’s gravity balances the Sun’s tidal pull, was 0.3576 au. 3I’s closest approach was 0.3580 au by our computation and 0.358330 au by JPL: 1.0011 and 1.0020 Hill radii. With the Hill radius at Jupiter’s mean distance (0.3552 au) the ratio is 1.008. The object that arrived from 28° from the Galactic Centre touched the balance sphere of the Sun–Jupiter pair and left (Fig. 4c). The approach was noted in advance by Loeb, Hibberd and Crawl, who showed that Juno could have been sent to within about 27 million km of the object [85].

The contact point.

- *When and how far:* 2026-03-16 11:33 UT (JPL: 12:22 UT), 53.56 million km from Jupiter, passing at 65.90 km/s relative to it.
- *Where in the Solar System* (heliocentric ecliptic J2000): $(-1.96319, +4.70277, -0.21433)$ au; longitude 112.658° , latitude -2.408° , 5.1006 au from the Sun.
- *Where in Earth’s sky:* RA $06^{\text{h}}50.6^{\text{m}}$, Dec $+20.32^\circ$, in Gemini, 4.731 au away and 4.14° from Jupiter (RA $07^{\text{h}}04.3^{\text{m}}$, Dec $+22.98^\circ$); Galactic longitude 194.60° , latitude $+8.93^\circ$, 17.1° from the Galactic anticentre. The object that arrived from near the Galactic Centre met Jupiter’s sphere, as seen from Earth, near the opposite point of the sky.
- *Where on Jupiter’s sphere:* 114° from the anti-Sun direction and 66° from the Sun direction, 42° below Jupiter’s orbital plane.
- *Relation to the throats:* the contact distance lies between Jupiter’s L_1 distance (0.3493 au,

sunward) and its L_2 distance (0.3655 au, anti-sunward), on the shell where the Sun–Jupiter balance sits; the point is 0.38 au from L_1 itself and 0.61 au from L_2 .

3I’s Jacobi constant was $C = -22.5$, against 3.0388 at L_1 and 3.0375 at L_2 : far below the throat energies, so the zero-velocity surfaces placed no constraint on its path. It crossed the surface of the Sun–Jupiter balance sphere at speed, where the slow bodies of Section 3.3 enter and leave only through the throats.

The configuration at the contact. At the contact Jupiter led Saturn by 111.049° in heliocentric longitude; in three dimensions the angle is 111.044° (latitudes $+0.332^\circ$ and -2.344°). The angle was growing by 0.0481° per day and passed two exact values within months of the contact (Fig. 2d): $\arccos(-\frac{1}{3}) = 109.4712^\circ$, the angle whose cosine is minus one third, on 2026-02-11 (32.6 days before), and 120° , one third of a turn, on 2026-09-23; an ephemeris error of 0.05° moves these dates by 1.0 day. The value 111.049° recurs twice per synodic cycle, with Jupiter and Saturn leading in turn: 2007-07-20, 2014-07-12, 2026-03-16, 2034-08-22. At the contact Saturn stood 8.00° from the Sun in Earth’s sky, and its conjunction with the Sun followed on 2026-03-25, 8.8 days later. Three further angles of the configuration at the contact were Saturn–Sun–3I 108.474° , Sun–Earth–Jupiter 109.235° and Sun–3I–Jupiter 110.658° . Around Saturn the sequence ran: equinox on 6 May 2025 [4]; 3I’s closest approach to Saturn, 9.9791 au, on 2025-05-28, 22.5 days later; 3I’s discovery on 1 July 2025; the southern decagon complete in August–September 2025; the contact on 16 March 2026. A body of 10 km radius and density 1000 kg/m^3 at 9.9791 au pulls on Saturn with $1.25 \times 10^{-19} \text{ m/s}^2$, 1.3×10^{-12} of the accelerations in Cassini’s residuals: the sequence is one of timing, and the signal in Cassini’s tracking is Saturn’s own.

X-rays of 3I. XRISM observed 3I for 17 hours on 26–28 November 2025 and found carbon, nitrogen and oxygen lines from solar-wind charge exchange extending about 400 000 km from the nucleus; XMM-Newton also saw the diffuse glow [35].

7.3 Radio: the Wow! signal, hydrogen and hydroxyl

The Wow! signal was recorded by the Big Ear telescope on 16 August 1977 at 03:16:06 UTC. The revised values are $1420.726 \pm 0.005 \text{ MHz}$, at least 73.4 s long, signal-to-noise ratio 30.1, at least 256 Jy, heliocentric radial velocity $-84 \pm 1 \text{ km/s}$ and LSR velocity $-74 \pm 2 \text{ km/s}$; the two horn positions are RA $19^{\text{h}}25^{\text{m}}02^{\text{s}}$ and $19^{\text{h}}27^{\text{m}}55^{\text{s}}$ ($\pm 3 \text{ s}$), Dec $-26^\circ57'18''$ and $-26^\circ57'13''$ ($\pm 20''$), and the beam is about $8'$ wide in right ascension [97]. The printout’s six characters 6EQUJ5 give the signal-to-noise ratio in successive 12-s samples, 6, 14, 26, 30, 19 and 5 (lower edges) [33]. The proposed natural cause is a maser-like flare of a small cold hydrogen cloud [96].

Direction. 3I came from RA 295.049° , Dec -19.072° (Galactic $l = 20.73^\circ$, $b = -19.06^\circ$), 8.619° from the positive-horn position and 8.371° from the negative-horn position; 1I and 2I lie more than 61° away (Fig. 5a). Loeb noted that 3I arrived from near the direction of the Wow! signal [84]. Its position and radial velocity in 1977 decide whether it could have been the source.

Position in 1977. Propagated back to the Wow! epoch, 3I was 591.12 au from the Sun, at RA $19^{\text{h}}39^{\text{m}}20.2^{\text{s}}$, Dec $-19^\circ05'30''$ (J2000), 0.204° from its incoming direction and 8.288° from the nearer horn. The offsets are $+152.7'$ in RA (times $\cos \text{Dec}$) and $+471.7'$ in Dec, against beam half-widths of $4'$ and $20'$.

Frequency. In August 1977 3I moved at 58.013 km/s, and its heliocentric radial velocity along the line of sight was -58.012 km/s, against the Wow!’s -84 ± 1 km/s, a difference of 25.988 km/s. Subtracting the observer’s velocity along the line (Earth’s orbital motion -15.000 km/s, its rotation $+0.026$ km/s at the Big Ear site) gives a topocentric velocity of -43.038 km/s. A transmitter on 3I at the hydrogen frequency would have been received at 1420.6097 MHz, 116.3 kHz below the Wow!, or 11.6 of the receiver’s 10-kHz channels; to appear at 1420.726 MHz it would have had to transmit at 1420.5221 MHz. Our ephemeris converts the Wow!’s 1420.726 MHz into -84.47 km/s heliocentric, against the published -84 ± 1 . The LSR radial velocities tell the same story: -46.45 km/s for 3I and -74 ± 2 km/s for the Wow!. 3I/ATLAS was not the source of the Wow! signal: in 1977 it lay 8.3° from the nearer beam centre, far outside a beam of half-widths $4'$ and $20'$, and a hydrogen-line transmitter on it would have been received 11.6 channels from the Wow!.

Hydrogen: a triad and a fourth. The Wow! lies 320.2 kHz above the hydrogen line, 1420.405751768 MHz [53], the flip between the two spin arrangements of hydrogen’s electron and proton. The upper level, $F = 1$, has three states, $m_F = -1, 0, +1$; the lower level, $F = 0$, has one. The line joins a triad to a single fourth state; the triad holds $\frac{3}{4}$ of the degeneracy. This is the decomposition $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$ whose singlet is the fourth state of the physics and quantum companions [135, 136].

Hydroxyl: two pairs about one centre. 3I’s radio lines are those of hydroxyl (OH). MeerKAT detected no OH on 20 and 28 September 2025, absorption at 1665 and 1667 MHz on 24 October and again on 4 and 6 November, and emission on 11–12 November [148, 3, 117]. Searches for artificial signals found none: down to 0.17 W over 900–1670 MHz with MeerKAT [148], 10–110 W over 1–9 GHz with the Allen Telescope Array [131], and about 0.1 W with the Green Bank Telescope [14]. OH’s ground level is a pair of levels of opposite parity (the Λ -doublet), and each member splits into $F = 1$ (three states) and $F = 2$ (five). The four lines join the upper member to the lower [70]: 1612 MHz is $F = 1 \rightarrow 2$, 1720 MHz is $F = 2 \rightarrow 1$, 1665 MHz is $F = 1 \rightarrow 1$ and 1667 MHz is $F = 2 \rightarrow 2$. Writing u_F and ℓ_F for the energies of the upper and lower members, $\nu_{1612} + \nu_{1720} = (u_1 - \ell_2) + (u_2 - \ell_1)$ and $\nu_{1665} + \nu_{1667} = (u_1 - \ell_1) + (u_2 - \ell_2)$ are the same combination of four level energies, so the two sums are equal. The measured frequencies satisfy this to the last digit:

$$1612.2309 + 1720.5299 = 1665.4018 + 1667.3590 = 3332.7608 \text{ MHz.}$$

Both pairs share one centre, 1666.38040 MHz. The main lines sit at ± 0.97860 MHz from it and the satellites at ± 54.14950 MHz: two pairs $\pm s$ about one zero (Fig. 5c). The hyperfine spacings are 55.1281 MHz in the upper member and 53.1709 MHz in the lower. The band between the hydrogen line and the OH lines is the “water hole”, named by Bernard Oliver in 1971, where searches for other civilisations have long listened [158]. The Wow! lies at its lower wall and 3I’s OH lines at its upper wall; the span from 1420.405752 to 1667.3590 MHz is a ratio of 1.17386, 2.775 semitones.

3I’s OH crossed its balance point. The OH lines of a comet appear in absorption or emission against the sky according to the sign of the population difference between the two members of the doublet. That sign is set by sunlight: the Doppler shift of the comet against the Sun’s absorption lines near $3086 \text{ \AA}$ changes how strongly each level is pumped (the Swings effect [29, 127], as used in [70]). Where the inversion passes through zero the two members are equally populated and the line vanishes. From our ephemeris, 3I’s heliocentric radial velocity on the MeerKAT dates was -45.12 and -40.13 km/s on 20 and 28 September (no OH), -8.452 km/s on 24 October (absorption), $+10.139$ and $+13.373$ km/s on 4 and 6 November (absorption),

and +20.929 and +22.332 km/s on 11 and 12 November (emission); the flip lies between +13.37 and +20.93 km/s. Comet C/2025 A6, observed by FAST in the same weeks while approaching the Sun, showed the opposite sequence [70]: emission at -22.39 and -19.85 km/s, absorption at -9.34 , -7.65 and -5.94 km/s, and no detection closer to zero; its flip lies between -19.85 and -9.34 km/s (Fig. 5d). Two comets, one on each side of zero velocity, crossed an inversion zero at speeds of similar size, as the Swings effect allows; they were at different distances from the Sun (0.54–0.65 au for C/2025 A6, 1.3687–1.4500 au for 3I).

7.4 The sounds of the giants and the visitors

Each source has a sound it is known for and a vibration of its own body (Table 20). Jupiter’s auroral electrons radiate by the cyclotron-maser process from a few kHz up to 40 MHz, a limit set by a surface field of up to 14 G [86, 37]; Saturn’s kilometric radiation lies near 100 kHz [37]; 3I’s OH line and rotation are as above, with a post-perihelion period of 7.136 ± 0.001 h [125]. As a terrestrial reference we include the periodical cicadas: *Magicicada septendecim* sings a modulated pure tone sharply tuned at 1.3 kHz, *M. cassini* in pulses with a broad peak near 6 kHz [168], and a cicada’s wings beat at 47 ± 1 Hz [152]. Periodical cicadas emerge every 13 or 17 years, both prime, so a 13-year and a 17-year brood meet only every $\text{lcm}(13, 17) = 221$ years; Broods XIX and XIII co-emerged in 2024 for the first time since 1803 and will next do so in 2245 [154].

source	A: sound	B: vibration or spin
cicadas	song 1300 Hz	wingbeat 47 Hz
Jupiter	radio up to 40 MHz	rotation 9.9250 h
Saturn	kilometric radio ~ 100 kHz	rotation $10^{\text{h}}33^{\text{m}}38^{\text{s}}$
3I/ATLAS	OH 1667.3590 MHz	rotation 16.16 h
Wow!	1420.726 MHz	–

Table 20: Characteristic frequencies of the sources: A, a sound each is known for; B, a vibration or spin of its body. Figure 5b adds further characteristic frequencies of the same sources.

Sound mappings. Each source is carried into hearing by whole octaves; the tones are synthesised from the published frequencies. The cicada song sounds at its own pitch, 1300 Hz, and the wingbeat two octaves up at 188 Hz. Jupiter’s spin sounds at 117.389 Hz (22 octaves up), its radio edge at 152.588 Hz and its acoustic spacing at 162.844 Hz. Saturn’s interior spin sounds at 110.324 Hz and beats against the hexagon’s 109.332 Hz at 0.9922 Hz, and its kilometric radio at 195.312 Hz. 3I’s OH pair sounds at 794.125 and 795.059 Hz (21 octaves down), beating at 0.9333 Hz, once every 1.072 s, and its spin at 144.194 Hz. The Wow! sounds at 677.455 Hz, shaped by 6EQUJ5 with each 12-s sample played as 1.2 s, over the hydrogen line at 677.302 Hz, with a beat every 6.549 s. Jupiter’s and Saturn’s spin tones stand 1.075 semitones apart. In each pair the beat is the difference of the two members: the OH beat is the main-line splitting 1.9572 MHz divided by 2^{21} , and the Wow! beat is its 320.2 kHz offset from hydrogen divided by 2^{21} .

8 A compact centre: the planets of PSR B1257+12

The first planets found beyond the Sun were found by arithmetic on pulse arrival times [163]. In the timing solution of Konacki and Wolszczan [74], with an assumed $1.4 M_{\odot}$ pulsar spinning every 6.21853 ms, planet A has $0.020 M_{\oplus}$ and a 25.262-day period; B has $4.3 M_{\oplus}$, 66.5419 days, $e = 0.0186$ and $i = 53^{\circ}$ or 127° ; C has $3.9 M_{\oplus}$, 98.2114 days, $e = 0.0252$ and $i = 47^{\circ}$

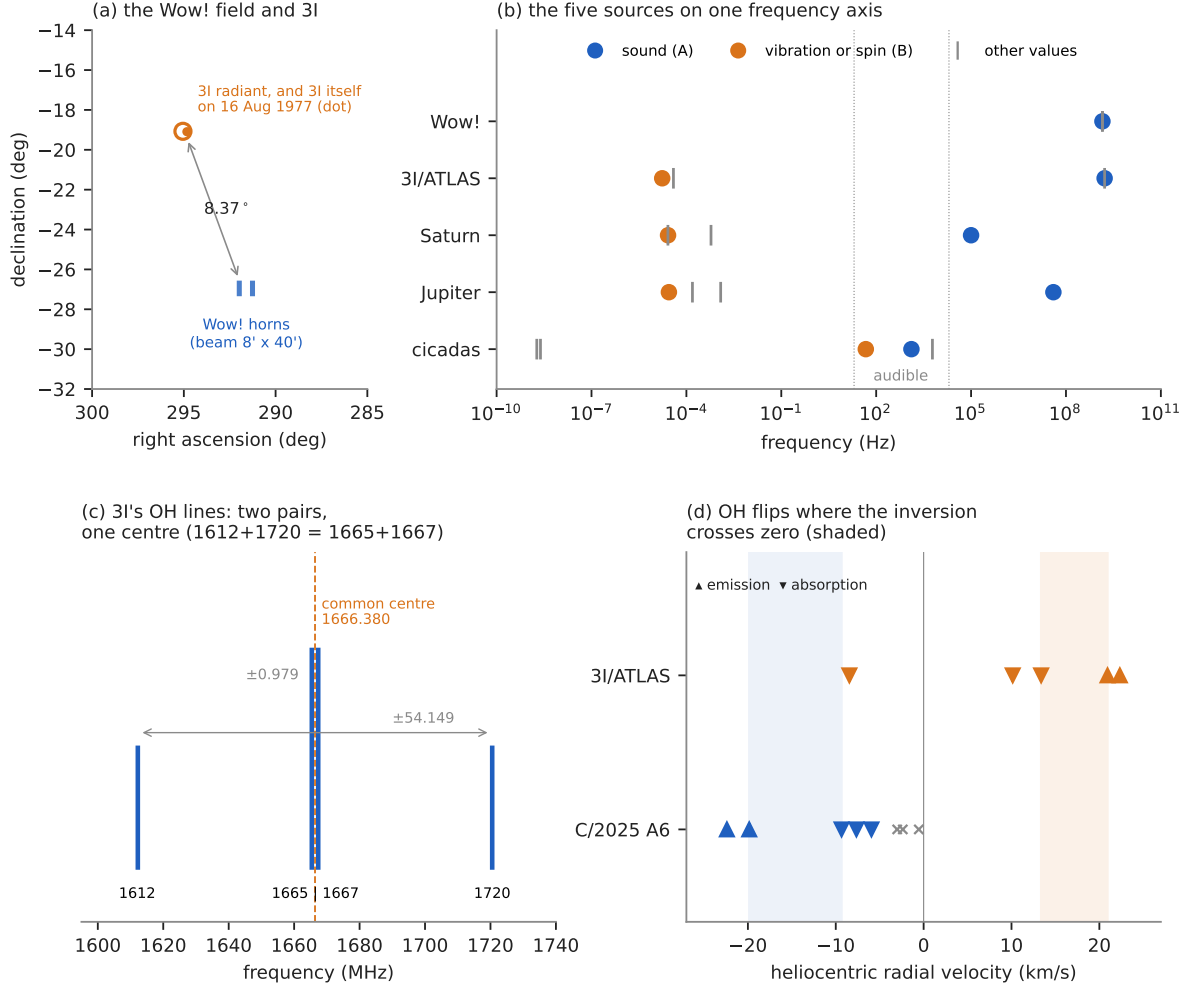


Figure 5: (a) The two Wow! horn positions with the beam footprint, and 3I's incoming direction with its 1977 position. (b) The five sources on one frequency axis: sounds (A), vibrations or spins (B) and further characteristic frequencies (grey). (c) The OH quartet: two pairs about one centre. (d) OH emission and absorption against heliocentric radial velocity for 3I/ATLAS (MeerKAT reports) and C/2025 A6 (FAST [70]); shading marks where each flip lies.

or 133° . Kepler's law puts them at 0.18849, 0.35951 and 0.46604 au. The pulsar's reflex orbits, $a_{\text{psr}} = a m / (M + m) = 496.13$ and 583.32 km for B and C, set against the projected semimajor axes $x = a_{\text{psr}} \sin i / c$ of the timing solution (1.3106 and 1.4134 light-ms), return $\sin i$ and $i = 52.4^\circ$ and 46.6° .

The pair. B and C lie near the 3:2 resonance: $P_C/P_B = 1.475933$, 1.60% inside it, so the frequency ratio is 0.6775 against $\frac{2}{3}$. The near-resonance repeats its geometry every $1/|3/P_C - 2/P_B| = 5.5863$ years, and it is the planets' mutual pull in this near-resonance, seen in the timing, that fixed their true masses and inclinations [74]. Timing cannot decide the mirror, i or $180^\circ - i$: the two solutions are reflections through the plane of the sky, the observer's horizon. B and C carry 0.4912 and 0.5072 of the system's orbital angular momentum, and A carries 0.0017. The ratio $L_C/L_B = (m_C/m_B)(P_C/P_B)^{1/3} = 1.0325 \pm 0.0715$ lies within 0.45σ of 1, treating the $\pm 0.2 M_\oplus$ mass errors as independent although they were fitted jointly. The pair is 14.31 mutual Hill radii apart, beyond the $2\sqrt{3} = 3.464$ that two planets on circular orbits need to avoid close encounters [44]; each carries the giants' two throats (Table 3), unequal by $h/3$ to 10^{-5} , and its Trojans would librate in 23.09 and 35.78 years.

The centre. A neutron star has no horizon, but its mass sets the same radii. For a non-rotating mass M in Schwarzschild coordinates ($G = c = 1$), light moves in the effective potential $V_\gamma = (1 - 2M/r)/r^2$, whose maximum $dV_\gamma/dr = -2/r^3 + 6M/r^4 = 0$ is the photon sphere $r = 3M$; massive bodies move in $V = (1 - 2M/r)(1 + L^2/r^2)$, whose circular orbits have $L^2 = Mr^2/(r - 3M)$ and are stable exactly for $r \geq 6M$, the innermost stable circular orbit [104]. The horizon lies at $2M$. Hence the horizon is exactly $\frac{1}{3}$ of the innermost stable orbit and $\frac{2}{3}$ of the photon sphere. A static observer sees circular orbits move at $v = [M/(r - 2M)]^{1/2}$, exactly $c/2$ at the innermost stable orbit and c at the photon sphere. For $1.4 M_\odot$ the three radii are 4.1346, 6.2018 and 12.4037 km, and the orbital frequency at the innermost stable orbit, seen from far away, is 1570.4 Hz. A neutron-star surface at 10–12 km lies between the photon sphere and the innermost stable orbit, with compactness $2GM/Rc^2$ of 0.41–0.34 (Fig. 1d); the pulsar’s light cylinder is at $cP/2\pi = 296.708$ km.

The observer inside the observation. Planet A’s 25.3-day period is close to the Sun’s rotation, and Scherer and co-workers proposed that solar rotation mimicked it [126]; Wolszczan and co-workers weighed a planet against a heliospheric propagation effect [164], and the 2003 solution keeps A as a planet [74]. A proposed fourth body was withdrawn when its signal was traced to slow changes in the dispersion measure [165]. What separates the cases is an equivalence: a companion’s pull delays the pulses equally at every radio frequency, while plasma, including the observer’s own solar wind, delays them as f^{-2} . One proposed origin of the system is a disk from the merger of two white dwarfs [118]: a pair merged into a centre, with planets forming around it.

9 Discussion: the four giants as balanced pairs

Table 21 collects the structure level by level. At every level a pair sits about a centre and balances at a zero that is either computed exactly or measured; at every level a third element or a fourth body enters in a stated way.

level	pair	balance (zero)	triad or fourth element
throats	L_1, L_2 about the planet	Hill radius; $\mathcal{A} = \frac{h}{3}(1 - \frac{h^2}{27} + \dots)$; transmission $T = \frac{1}{2}$ at the saddle	$\lambda^2 - \omega^2 = 2, \lambda^2\omega^2 = 3^3$; 3I crossed the sphere at $1.0011 r_H$
hearts	the two hemispheres; Jupiter and Saturn	winds end at 10^5 bar in both	ring of N with a centre; $3 + 1$ exactly on the edge
gas ladder	NH_3 and H_2S	$N = S$: neither survives	rungs 3, 3, 4, 4: the triad of decks, and CH_4 as a fourth
the set	Jupiter and Saturn	conjunctions at $\frac{2}{3}$ of a turn	the trigon; Io, Europa, Ganymede and Callisto; three live modes and one zero mode
triangle	two pairs of opposite Krein sign	collision at $\mu_G, \omega = 1/\sqrt{2}$	a triad stands only around a dominant centre
radio	OH main and satellite pairs	common centre 1666.38040 MHz	hydrogen: the triplet and the singlet
visitors	1I, 2I, 3I	no common direction or origin	the Sun as common focus, placed by the observer
PSR B1257+12	planets B and C	$L_C/L_B = 1.03 \pm 0.07$	horizon $\frac{1}{3}$ of the ISCO; planet A as the small third

Table 21: The four giants and their visitors, level by level.

Outlook. We predict undiscovered retrograde irregular moons of Uranus with apocentres of 0.60–0.73 of its Hill radius, 42–51 million km from the planet (Section 6.7), which a deep survey of that annulus will find, and an acoustic spacing of Saturn of 112–117 μHz (Section 4.4), which a long continuous record of Saturn’s disk or gravity will measure. A deep N/S for Neptune, from an entry probe or microwave sounding, will test Theorem 3 at the second ice giant, and a measurement of the ortho/para state of hydrogen will remove the largest model uncertainty of the ice-giant ladder (up to 78% in the water base) and, with O/H, place its water rung.

10 Data and sources

- *Planetary constants:* NASA NSSDC planetary fact sheets for Jupiter, Saturn, Uranus and Neptune (D. R. Williams; updated 2 October 2024, 18 March 2025, 2 October 2024 and 3 October 2024) [159, 161, 162, 160]: GM (126.687, 37.931, 5.7940, $6.8351 \times 10^6 \text{ km}^3\text{s}^{-2}$), semimajor axes (778.479, 1432.041, 2867.043, $4514.953 \times 10^6 \text{ km}$), sidereal periods (4332.589 and 10759.22 days for Jupiter and Saturn), rotation (Jupiter 9.9250 h, Neptune 16.11 h), equatorial radii of Uranus and Neptune (25559, 24764 km), J_2 of Uranus and Neptune (3343.43 , 3411×10^{-6}), obliquities, dipole tilts and offsets, 1-bar and 0.1-bar temperatures, 1-bar gravities and bulk compositions (Tables 2 and 4); Jovian and Saturnian satellite fact sheets for the moon periods [109, 110]. $GM_\odot = 1.32712440018 \times 10^{20} \text{ m}^3\text{s}^{-2}$, within 2×10^{-10} of the IAU 2009 value, and 1 au = 149597870.7 km [87]; the planetary mass ratios $M_\odot/m$ (1047.348644 Jupiter, 3497.9018 Saturn, 3098703.59 Mars, 332946.0487 Earth) and the Gaussian constant 0.01720209895 used for the encounters [87].
- *Gravity fields and interiors:* Jupiter’s J_2, J_4, J_6 [31]; Saturn’s J_2, J_4, J_6 and the Grand Finale residuals [60]; interior models and heavy-element masses [151, 101, 92]; wind depths [72, 73, 41]; p-mode signals [32, 93, 43]; Uranus’s rotation 17.247864 h [76].
- *Ring seismology:* Saturn’s seismic rotation period, $10^{\text{h}}33^{\text{m}}38^{\text{s}}$ with its 5% and 95% quantiles, and the counts of density and bending waves [91]; the stably stratified core [92].
- *Poles:* Jupiter’s cyclones [1, 108, 106, 82, 18]; Saturn’s hexagon, decagon, equinox, radio periods and dipole tilt [122, 123, 147, 4, 39, 16]; Jupiter’s X-ray aurorae [45, 30].
- *Gas ladder:* all inputs of Table 4, with the compilations [5, 6] and the observations [166, 80, 81, 49, 50, 40, 141, 105, 61, 62, 63, 25]; the published cloud bases of Table 7; secondary values quoted through these papers: the Galileo temperature profile (Seiff et al. 1998), the Galileo cloud base (Banfield et al. 1998) and the NH_3 -ice coverage (Baines et al. 2002), all as quoted in [6, 50], and the ice-giant CH_4 of Karkoschka and Tomasko, as stated in [63]. Thermodynamic data: IAPWS equations for water [150, 149]; the NIST Chemistry WebBook [83] for the NH_3 , H_2S and CH_4 triple points, sublimation enthalpies and Antoine equations, compiled there from Timmermans (1921), Fonseca and Lobo (1989), Overstreet and Giauque (1937), Stull (1947), Goodwin (1983), Clark et al. (1951), Giauque and Blue (1936), Stephenson and Malanowski (1987) and Prydz and Goodwin (1972); H_2 spectroscopic constants [55] and heat capacities [78] via the WebBook; the NH_4SH constant [6]; physical constants CODATA 2018 [19].
- *Sensor ladder:* the UCI Air Quality data set (hourly responses of five metal-oxide sensors with reference analysers, 10 March 2004 to 4 April 2005, 9357 rows) [27, 26], as analysed in the chemistry companion [137]; used here only in Fig. 3d and Section 5.9.
- *Interstellar objects:* orbital elements of 1I (JPL Small-Body Database, solution 16: $e = 1.201133796$, $q_p = 0.2559115813 \text{ au}$, $i = 122.7417^\circ$, $\Omega = 24.5969^\circ$, $\omega = 241.8105^\circ$) [68], 2I ($e = 3.3565$, $q_p = 2.00652 \text{ au}$, $i = 44.053^\circ$, $\Omega = 308.15^\circ$, $\omega = 209.12^\circ$, epoch JD 2458853.5) [156] and 3I ($e = 6.14135$, $q_p = 1.35645 \text{ au}$, $i = 175.12^\circ$, $\Omega = 322.17^\circ$, $\omega = 128.02^\circ$, epoch JD 2461090.5) [157]; heliocentric velocities [89, 142, 24]; solar motion [129]; all other published properties with their sources in Tables 13 to 17; JPL’s 3I–Jupiter approach, 0.358330 au at

- 12:22 UT on 16 March 2026 [157]; X-ray observations of 3I [35]; PSO J318.5–22 [11].
- *Radio*: the Wow! parameters and horn positions (Table 4 of [97]); the printout code [33]; the hydrogen hyperfine frequency 1420405751.768 ± 0.002 Hz [53]; the OH ground-state frequencies (Table 1 of [70]); the C/2025 A6 radial velocities and heliocentric distances (Tables 2–3 of [70]); MeerKAT dates and results [148, 3, 117]; technosignature limits [148, 131, 14]; the Big Ear site, latitude 40.25° N, longitude 83.05° W (Earth-rotation velocity only, at most 0.36 km/s).
 - *Sources of sound*: Jupiter’s and Saturn’s radio [86, 37]; 3I’s rotation [124, 125]; cicada songs, wingbeat and broods [168, 152, 154].
 - *Dynamics*: Ohtsuka’s captures [113]; Trojan populations [155, 23, 57, 111]; moon lists of Jupiter (Wikipedia group tables: 13 outer moons with eccentricities, 21 with semimajor axes only), Saturn (210 Norse-group moons with eccentricities, 35 Gallic and Inuit moons with semimajor axes only), Uranus (29) and Neptune (16), read 24 September 2026 [155]; irregular-satellite stability limits [112].
 - *PSR B1257+12*: the timing solution of Konacki and Wolszczan [74]: pulse period 6.21853194840048 ms; A: $x = 0.0030$ ms, $P = 25.262$ d, $0.020 M_\oplus$; B: $x = 1.3106$ ms, $e = 0.0186$, $P = 66.5419$ d, $\omega = 250.4^\circ$, $4.3 \pm 0.2 M_\oplus$, $i = 53^\circ$ or 127° ; C: $x = 1.4134$ ms, $e = 0.0252$, $P = 98.2114$ d, $\omega = 108.3^\circ$, $3.9 \pm 0.2 M_\oplus$, $i = 47^\circ$ or 133° ; pulsar mass $1.4 M_\odot$ assumed; $GM_\odot/c^2 = 1476.625$ m.
 - *Ephemeris*: planetary positions from Schlyter’s method [128], rotated to the J2000 ecliptic, validated against JPL’s 3I–Jupiter, 3I–Mars and 3I–Earth approaches (Section 7.2).
 - *Formulas*: Hill’s problem and the restricted problem [107]; the neck topology [20, 94, 75, 46]; the flux theorem [88]; the parabolic barrier [77]; the Gascheau–Routh criterion and Krein’s theorem [42, 121, 98]; the Laplace–Lagrange secular theory [107]; the Radau–Darwin relation [100]; the $N + 1$ vortex problem [15, 9]; the Schwarzschild radii [104]; the two-planet stability limit [44].

All other numbers are computed from these inputs by the formulas and methods stated where they appear.

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